https://pubsonline.informs.org/journal/mnsc

MANAGEMENT SCIENCE
Vol. 68, No. 10, October 2022, pp. 7298–7322
ISSN 0025-1909 (print), ISSN 1526-5501 (online)

# Data Breach Announcements and Stock Market Reactions: A Matter of Timing?

**Jens Foerderer,[a,*] Sebastian W. Schuetz[b]**

[a] Department of Innovation and Entrepreneurship, TUM School of Management, Technical University of Munich, 80333 München, Germany; [b] Department of Information Systems and Business Analytics, College of Business, Florida International University, Miami, Florida 33199
*Corresponding author

**Contact:** jens.foerderer@tum.de, https://orcid.org/0000-0002-3090-4559 (JF); sebastian.w.schuetz@outlook.com, https://orcid.org/0000-0003-0503-4472 (SWS)

**Received:** September 30, 2019
**Revised:** October 9, 2020; March 30, 2021; July 30, 2021
**Accepted:** September 7, 2021
**Published Online in Articles in Advance:** February 15, 2022

**https://doi.org/10.1287/mnsc.2021.4264**

**Copyright:** © 2022 INFORMS

**Abstract.** Although firms' announcement of data breaches can lead to reputational or operational damages, extant research suggests that stock markets are relatively unresponsive to such announcements. We investigate whether markets' unresponsiveness can be explained by firms strategically timing the announcement to coincide with busy days in the media, thereby reducing attention and, ultimately, attenuating market reactions. We leverage novel data on data breach announcements in the United States between 2008 and 2018 and create a measure of busyness in the trade press—news pressure—based on the *Wall Street Journal*. To investigate, we conduct two complementary studies. In Study 1, we employ an instrumental variable approach to assess whether announcements coincide with days of predictably high news pressure. We find that this is the case. On days with a one-standard-deviation-higher predictable news pressure, 4.44% more data breaches are announced (or approximately 19.024 data records). Strategic timing is more prevalent for breaches that are severe, that have firm-internal causes, and that leak healthcare data or credentials. In Study 2, we utilize a stock market event study to assess market reactions conditional on news pressure on the announcement day. We find that data breach announcements are associated with negative market reactions, yet these are attenuated by higher news pressure on the announcement day. If news pressure is on its empirical mean (respectively, one standard deviation above), we estimate a median decline in market capitalization of $347 (respectively, $85) million. We conclude that firms' strategic timing might explain inconsistent findings in prior work.

**History:** Accepted by Chris Forman, information systems.
**Supplemental Material:** Data and the online appendix are available at https://doi.org/10.1287/mnsc.2021.4264.

**Keywords:** information systems privacy • data breaches • news pressure • event study • media attention

## 1. Introduction

Data breaches are of significant interest to researchers in information systems and management (Miller and Tucker 2009, Romanosky et al. 2011, Acquisti et al. 2015). Data breaches are security incidents in which personally identifiable information, such as Social Security numbers, driver's license numbers, names, and birthdays, is exposed to unauthorized third parties (Romanosky et al. 2011). Every year, over a thousand U.S. firms report data breaches encompassing hundreds of millions of consumer records (Identity Theft Resource Center (ITRC) 2020). Whereas consumers worry that their personal information can be abused for identity theft, shareholders' concerns relate to reputational damages, operational disruptions, or litigation costs (see Romanosky et al. 2014, Goode et al. 2017, and Martin et al. 2017). In light of these potential consequences, one would expect negative stock market reactions in response to firms' data breach announcements.

Despite these consequences, extant studies suggest that markets appear to be relatively—and perhaps surprisingly—unresponsive to data breach announcements. Kvochko and Pant (2015) provide several examples of firms that suffered significant data breaches but not persistent declines in their stock prices. Systematic empirical assessments of market reactions to data breach announcements also yield inconsistent findings. Whereas several studies report negative market reactions (e.g., Acquisti et al. 2006, Malhotra and Malhotra 2011, and Martin et al. 2017), others document null, small, or short-lived effects (e.g., Campbell et al. 2003, Kannan et al. 2007, and Goel and Shawky 2009).

In this paper, we develop and test one potential explanation for markets' unresponsiveness to data breach announcements—namely, that firms strategically time breach announcements, which can eventually attenuate investor reactions. The notion of strategic timing

has been explored in the economics literature and refers to the act of releasing announcements on days when the news media are already busy covering other newsworthy events (Eisensee and Strömberg 2007, Garz and Sörensen 2017, Durante and Zhuravskaya 2018). In this study, we test whether this strategy is used by firms to reduce media and public attention toward the breach announcement and, ultimately, explains observed attenuated stock market reactions. From a theoretical perspective, this argument derives from the notion that the news media play a critical role in shaping public attention and sentiment (McCombs and Shaw 1972, Caroll and McCombs 2003). Media space and time are finite, and editorial boards face pressure to attract their audiences' attention by publishing the most important and relevant news. If news media are occupied, data breach announcements might be crowded out from reporting.[1] Drawing on Eisensee and Strömberg (2007), we refer to the busyness of the news media on a given day as news pressure. Strategic timing also offers a feasible explanation. To protect consumer data, U.S. federal and state legislatures have passed regulations that mandate firms to publicly disclose data breaches, but they provide firms some flexibility on when to make the announcement (Schwartz and Janger 2007).

An empirical investigation of this explanation is not straightforward because we cannot observe whether an individual firm strategically times an announcement. Firms do not disclose their use of this strategy. Even if we can observe the time between a data breach's discovery and its announcement, we would remain unable to determine whether the announcement was timed or whether it simply took a firm more time to investigate the breach.

To overcome these obstacles, we conduct two complementary analyses with the goal of indirectly testing the strategic timing explanation. In Study 1, we test whether days with predictably higher levels of news pressure also show a significantly higher number of data breach announcements. From an econometrical point of view, a mere correlation between news pressure and the occurrence of data breach announcements would be misleading because firms cannot time their actions to news pressure caused by unpredictable events. Therefore, we implement an instrumental variable design to isolate the portion of news pressure that can be foreseen by firms. In Study 2, we examine stock market reactions to data breach announcements given different levels of news pressure on the announcement day. We follow extant work and estimate cumulative abnormal returns in a stock market event study framework. Both studies rely on a novel data set of data breach announcements made between 2008 and 2018 by firms operating in the United States. We were provided with the data by the ITRC. We infer daily news pressure from the crowdedness of the *Wall Street Journal*'s What's News: Business & Finance column.

Study 1 provides consistent evidence that predictably busy days in the press show more data breach announcements, which we interpret as evidence of strategic timing. Calendar days with a one-standard-deviation-higher predictable news pressure show 4.44% more data breach announcements, representing 19.024 individual records leaked. A record refers to data about one person or individual (e.g., customer, patient, or employee). This observation is adjusted for day-of-week, week, month, and year fixed effects, as well as for lags and leads. This finding is corroborated with several robustness checks, including checks for sampling bias, exclusion of subsamples, weighting of the instrument, alternative operationalizations of news pressure, and placebo tests. Furthermore, we find that strategic timing is more prevalent for (1) severe than nonsevere data breaches, (2) data breaches of public than nonpublic firms, (3) data breaches that are due to firm-internal (i.e., employee errors, theft) rather than firm-external causes (i.e., hacking, ransomware), and (4) breaches of healthcare and financial data as well as credentials.

Study 2 documents consistent evidence that cumulative abnormal returns (CARs) on the days around the announcement (i.e., [−3, 5]) are negative and significant yet less negative the higher the news pressure on the day of the announcement. In terms of market capitalization impacts, if news pressure is on its empirical mean (respectively, one standard deviation above), we estimate that a data breach announcement is associated with a median decline in market capitalization by $347 (respectively, $85) million. This finding is robust to alternative event windows, expected return models, subsamples, model parameters, and time fixed effects. We further implement tests for pre-event heterogeneity, placebo events, and additional rival explanations. We conclude from Study 2 that negative market reactions to data breach announcements are attenuated by higher news pressure on the announcement day.

Taken together, we conclude that the inconsistent market reactions to data breach announcements could be partially explained by firms' strategic timing. Our findings contribute to several streams of research. By demonstrating that firms may attenuate market reactions through strategic timing, our study helps understand the causes of the observed unresponsiveness of markets to data breach announcements (e.g., Kvochko and Pant 2015 and Kwon and Johnson 2015) and help explain inconsistent findings in prior work (cf. Ko and Dorantes 2006, Kannan et al. 2007, and Martin et al. 2017). Furthermore, our work adds to research investigating firms' postbreach behaviors by providing evidence of strategic timing and helps in understanding

the incentives for firms to engage in strategic timing (e.g., Romanosky et al. 2011, Laube and Böhme 2016, and Goode et al. 2017). Our findings also offer rich insights into the factors that make strategic timing more likely. Finally, our study contributes to the broader discourse on firms' timing of bad news (e.g., DellaVigna and Pollet 2009, Michaely et al. 2014, and Michaely and Rubin 2016) by documenting another instance of firms timing based on news pressure and establishing downstream market consequences (e.g., Eisensee and Strömberg 2007 and Durante and Zhuravskaya 2018).

## 2. Theoretical Background

### 2.1. Data Breach Notification Laws in the United States

Beginning with the California Civil Code Section 1798.29 of 2003, all U.S. states have, as of 2021, implemented legislation that mandates firms to disclose the occurrence of data breaches (see Section 1 in Online Appendix A for an overview). The objectives of these data breach notifications laws are to empower affected individuals to take protective actions against possible identity theft and to create market incentives for firms to strengthen the security of consumer data by making data breaches known to the public (Romanosky et al. 2011, Bisogni 2016, Laube and Böhme 2016). The expectation is that public attention will produce negative consequences to incentivize firms to improve data security controls (Bisogni 2016). Romanosky et al. (2011) call this approach "sunlight as a disinfectant" (p. 262), with notifications shining "sunlight" on firms and thereby "disinfecting" them from poor security practices.

All such regulations mandate firms to disclose data breaches to authorities and notify affected consumers (IT Governance 2019). The specific disclosure requirements differ between states. First, the threshold at which data breaches require notification is, depending on the state, between 1 and 1,000 breached records. Second, regulations differ in the time frames imposed on firms to notify affected consumers. Regulations urge firms to make data breach announcements after the first notice "in the most expedient time possible" (i.e., California Civil Code 1798.29(a)) or "as soon as practicable" (i.e., Massachusetts General Laws, Part I, Title XV, Chapter 93H, Section 3). Some states have imposed a specific deadline between 30 and 90 days.[2]

### 2.2. Market Reactions to Data Breach Announcements

Whether data breach announcements engender negative stock market reactions is currently unclear, as prior examinations have yielded inconsistent results (see Section 2 in Online Appendix A for an overview). First, the results are inconsistent regarding the existence of negative market reactions. Whereas several studies document negative market reactions (e.g., Acquisti et al. 2006, Malhotra and Malhotra 2011, and Martin et al. 2017), a considerable number of other studies document null effects or limited evidence (e.g., Campbell et al. 2003, Kannan et al. 2007, and Goldstein et al. 2011). On the basis of a systematic review of extant event studies examining the impact of various types of information technology (IT) security breaches on stock market valuations published between 1998 and 2015, Spanos and Angelis (2016) report that only 71% of the reviewed studies could confirm a significant negative impact. Second, the results are inconsistent with regard to the persistence of the observed effects. Whereas a few studies find evidence that negative abnormal returns persist for several weeks (e.g., Gatzlaff and McCullough 2010 and Morse et al. 2011), others observe significant effects only in the immediate aftermath of the announcement (i.e., zero, one, or two days) but not beyond (e.g., Goel and Shawky 2009 and Martin et al. 2017). Third, among the studies that observe negative market reactions, the effect magnitudes differ. For instance, Cavusoglu et al. (2004) observe that data breach announcements lead to a 2.1% decline in cumulative abnormal returns within two days; however, Malhotra and Malhotra (2011) note a decline of only 0.78% over the same time window, and other studies' estimates are even lower (e.g., Martin et al. 2017).

So far, only a few papers have empirically investigated the causes of the inconsistencies (Malhotra and Malhotra 2011, Kvochko and Pant 2015, Martin et al. 2017). It is suggested that factors such as data breach severity as well as firm size and industry can influence market reactions. These insights notwithstanding, the role of announcement timing has not been investigated. Related is also research into firms' post-breach behaviors (Goode et al. 2017, Gwebu et al. 2018, Bansal 2019). Those studies documented the importance of recovery behaviors toward consumers, such as public apologies and consumer compensation campaigns, yet did not study announcement timing.

### 2.3. News Pressure and Strategic Timing

The role of news pressure has been investigated by nascent research in media economics. Extant work documents that, on days with high news pressure, the media can be limited in their capacity to cover all relevant events and thus crowd out coverage of some events, such as military actions, disaster relief, and criminal investigations of politicians (Eisensee and Strömberg 2007, Garz and Sörensen 2017, Durante and Zhuravskaya 2018). The notion of timing is also related to research in finance, where several studies have investigated whether firms release "bad" news

during periods when markets are supposedly less attentive, such as after trading hours or on Fridays, and whether such an announcement practice allows firms to reduce market penalties (e.g., Patell and Wolfson 1982, Damodaran 1989, Hong and Stein 1999, DellaVigna and Pollet 2009, and Doyle and Magilke 2009). However, for both the notion of timing and the effects on markets penalties, recent evidence is mixed (Michaely et al. 2014, deHaan et al. 2015, Michaely and Rubin 2016). Thus, it remains unclear whether strategic behavior regarding the announcement can influence stock market reactions. Our study draws on these works, but we are primarily interested in explaining inconsistent market reactions to data breach announcements as well as in understanding firms' incentives to engage in this behavior.

## 3. Theory and Hypotheses

### 3.1. Agenda-Setting Theory

We argue that strategic timing can attenuate stock market reactions by reducing media coverage and subsequent investor attention. To understand the effect of media coverage on investor attention, we turn to agenda-setting theory, which proposes a strong relationship between the issues that are covered by the news media and the public's attention and opinions (McCombs and Shaw 1972, 1993). The influence process begins with the coverage of events (Caroll and McCombs 2003). As the public has limited time and resources, they rely on the consumption of news media to acquire information about events. The news media's coverage provides cues that inform the audience of a particular news story's importance (e.g., title story on the TV news, front-page article in a newspaper). As the audience uses these cues to infer the importance of the story (Caroll and McCombs 2003), the most salient issues in the news media become the most salient issues to the audience.

### 3.2. Hypotheses

We argue that media coverage might substantially influence the market reaction to a data breach announcement. First, media coverage of a data breach announcement influences whether retail investors take notice of the event. Unlike professional investors and analysts who closely follow firms, retail investors spend less time on analysis and rely on different sources (Da et al. 2011). Thus, retail investors might overlook data breaches that do not receive extensive media coverage.

Second, media coverage of a data breach announcement influences public perceptions of the event. Because media coverage directs public attention and serves as a cue to a topic's salience (McCombs and Shaw 1972, 1993), less media coverage conveys the message that a data breach is of lower importance. If affected individuals interpret a data breach as less important, they might be less likely to sue the firm or express their outrage publicly. The media might further amplify pressures, invoking executive turnovers or compensations (Goode et al. 2017). As a consequence, the firm might suffer less from reputational damages or litigation cost. It is therefore plausible that also professional investors respond less negatively to data breach announcements that receive less media coverage. For these reasons, we expect firms to have strategic incentives to reduce media coverage.

A crucial determinant of media coverage is news pressure (Eisensee and Strömberg 2007). The influence of news pressure arises because media coverage itself depends on editorial decisions. Editorial boards are dedicated to reporting news that is deemed relevant and important to their audiences (McCombs and Shaw 1972, 1993). However, on any given day, news outlets are limited in their capacities to cover events. If the number of relevant events increases, media outlets may only be able to cover a subset—the most newsworthy—of them. Thus, events that occur on busy days might receive less coverage and, ultimately, less attention. We therefore expect that news pressure can crowd out data breach announcements, and thus firms will strategically time their data breach announcements to days of high news pressure.

Strategic timing is also feasible. First, news pressure is to a certain degree predictable (e.g., prescheduled quarterly earnings releases, sports and entertainment events, political events). Second, as outlined before, notification laws provide firms with leeway on making the announcements within a specific time frame after the occurrence of a data breach. Moreover, it is difficult to audit the exact time a data breach was discovered internally.

Taken together, and in light of the impossibility to observe strategic timing directly, we anticipate the following two observations. First, we expect the timing of data breach announcements to be correlated with predictable news pressure. Second, we expect data breach announcements that occur on days with high news pressure to be associated with less negative stock market reactions because on days with high news pressure, data breach announcements might be crowded out by other more newsworthy events. Thus, we offer the following hypotheses.

**Hypothesis 1.** *Higher predictable news pressure is associated with a higher number of data breach announcements.*

**Hypothesis 2.** *Higher predictable news pressure on the day of a data breach announcement is associated with less negative stock market reactions.*

# 4. Data

## 4.1. Data Breach Announcements

To test our hypotheses, we obtained a novel data set on data breach announcements from the ITRC, a nonprofit that tracks data breaches for entities that are located or conduct business in the United States and that affect U.S. consumers. The ITRC defines a data breach as "an incident in which an individual name plus a Social Security number, driver's license number, medical record or financial record (credit/debit cards included) is potentially put at risk because of exposure" (ITRC 2019, p. 19). This definition is relatively broad and, therefore, offers a large sample to consider.

The ITRC provided us with all recorded data breach announcements—8,886 in total—between January 1, 2008, and December 31, 2018. To ensure comparability, we only retained data breach announcements exceeding 1,000 breached records (see Section 3 in Online Appendix A for details on the filtering).

Table 1 presents the data breach announcements by cause, type of data leaked, and year. Most data breaches occurred in California (15.03%), followed by Texas (7.15%), New York (6.97%), and Florida (5.62%). The discovery date (i.e., when a firm became aware of a breach) is available only for a small fraction of the announcements. For data breaches where the discovery date is available, the average (respectively, median) number of days between the announcement of a breach and the date when firms reported becoming aware of the breach is 39 (respectively, 28). Only 10% of the breaches were announced within five days of their discovery.[3]

**Table 1.** Description of Data Breaches in the Sample

| Characteristic | *N* |
|---|---|
| Cause of breach | |
| Accidental exposure | 262 |
| Hacking | 1,125 |
| Insider theft | 228 |
| Unauthorized access | 244 |
| Phishing/skimming | 200 |
| Ransomware | 225 |
| Employee error | 287 |
| Not reported | 1,603 |
| Type of data | |
| Social Security number | 3,092 |
| Financial | 1,505 |
| Health | 1,677 |
| Credentials | 440 |
| Other | 2,691 |
| Announcements per year (by public firms) | |
| 2008 | 188 (71) |
| 2009 | 101 (44) |
| 2010 | 195 (64) |
| 2011 | 139 (41) |
| 2012 | 149 (37) |
| 2013 | 229 (50) |
| 2014 | 329 (65) |
| 2015 | 234 (77) |
| 2016 | 395 (98) |
| 2017 | 585 (97) |
| 2018 | 461 (107) |
| Announcements per firm (by public firms) | |
| 1 | 2,725 (571) |
| 2 | 93 (54) |
| 3 | 24 (4) |
| 4 | 4 (5) |
| 5 or more | 1 (5) |

*Notes.* This table reports descriptive statistics on the data breach announcements included in the sample. Please note that numbers on data breach causes and data types may not add up because of missing reports or multiple classifications.

## 4.2. News Pressure

The primary independent variable in our analyses is news pressure. We develop a measure of news pressure that is based on a key outlet of the business press—namely, the *Wall Street Journal* (*WSJ*). We define news pressure in the *WSJ* as the number of news items in the front-page column What's News: Business & Finance that are unrelated to data breach announcements.[4] A central element of the *WSJ* front page since the early 1930s, the What's News column represents a concise digest of the most newsworthy events from the perspective of the editors, along with a reference to the corresponding coverage within the issue. Each news item appears as a separate bullet point. The column occupies a relatively fixed amount of space limited by the length of the front page (see Section 5 in Online Appendix A for an exemplary front page). We argue that on days when several important events take place, more articles must be covered in the column (i.e., increasing the pressure on the column), which reduces the space available for other news stories, including data breach announcements.

We rely on the *WSJ* for three reasons. First, the *WSJ* is likely to capture relevant news pressure because business news represents an important part of its reporting (e.g., Kaniel and Parham 2017). Second, the *WSJ* has a broad reach beyond the business world. Among the six major U.S. newspapers by print circulation over the sample period, *WSJ* consistently ranked either first or second (Agility PR Solutions 2020). In addition, the *WSJ* is frequently cited by other media. Third, by relying on the *WSJ*, our study aligns with prior work that has documented the influence of the *WSJ* on business outcomes (e.g., Kaniel and Parham 2017).

We construct two measures of news pressure: an uncorrected measure (*NEWS PRESSURE UNCORRECTED*) and a corrected measure (*NEWS PRESSURE*).[5] The corrected measure seeks to overcome a mechanical correlation between data breach announcements and news pressure on days where the What's News column reports data breach incidents. The corrected measure

excludes articles that reported data breach announcements, whereas the uncorrected measure does not. When the column does not report any data breach announcement, both measures are equivalent. If the column reports a data breach announcement, the corrected news pressure measure is equal to the article count minus the number of data breach announcement items. Section 6 in Online Appendix A provides examples for the computation of both measures.

Figure 1 plots the seven-day moving average of (corrected) *NEWS PRESSURE*. There is considerable variation over time in the variable, with relative busyness during the Global Financial Crisis of 2008, the Europe/Greece crisis in 2013, and the 2016 U.S. elections. The column underwent one structural change in August 2013, when its design changed from a two-column format to a one-column format, which eventually reduced the space by half. We control for this change in the regressions.[6]

Figure 2 plots (logged) *NEWS PRESSURE* from January to July 2009, with summaries of the column items for the days with the highest and lowest news pressure for illustration. One of the busiest days in that period was February 25, 2009, when several newsworthy events took place, such as the Obama administration revealing plans to subject banks to stress tests, the Federal Reserve increasing its stake in Citigroup to 40%, and Steve Jobs disclosing his personal health issues.

We collected the daily What's News: Business & Finance column for the observation period from Factiva. The *WSJ* appears Mondays to Fridays and in a weekend edition on Saturdays. The weekend edition utilizes a different design; therefore we cannot compute the measure for Saturdays.[7] We also dropped U.S. public holidays from the sample (e.g., Labor Day, Independence Day). The effective sample thus covers nonholiday Mondays to Fridays, with news pressure on day $t$ inferred from the next morning's issue of the *WSJ*. News pressure on Friday is then inferred from the following Monday's issue. On some days, the column reports "Amplifications & Corrections" as the topmost item, which we do not count as a news item in the measure. We dropped days for which the Factiva records were corrupt or erroneous.

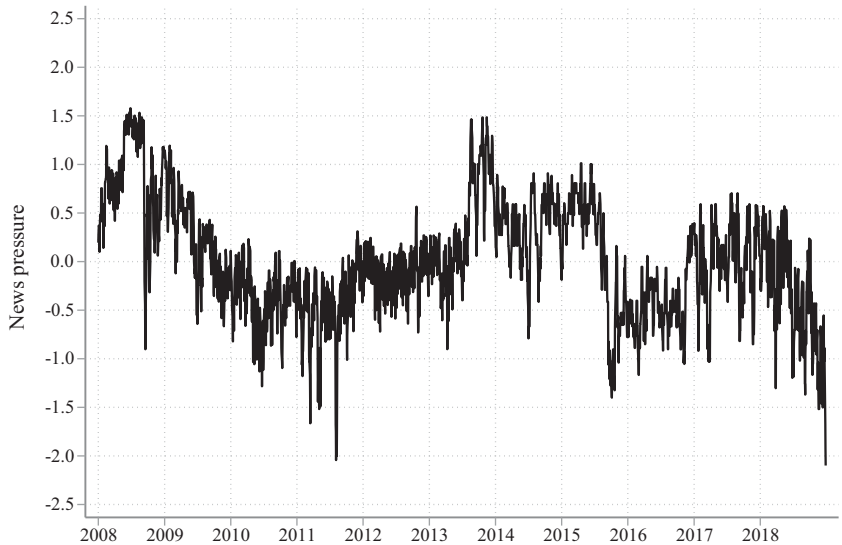
Table 2 provides an overview of the variables and corresponding data sources. The summary statistics are reported in Online Appendix A, Section 7.

## 5. Study 1: Strategic Timing of Data Breach Announcements (Hypothesis 1)

### 5.1. Research Design

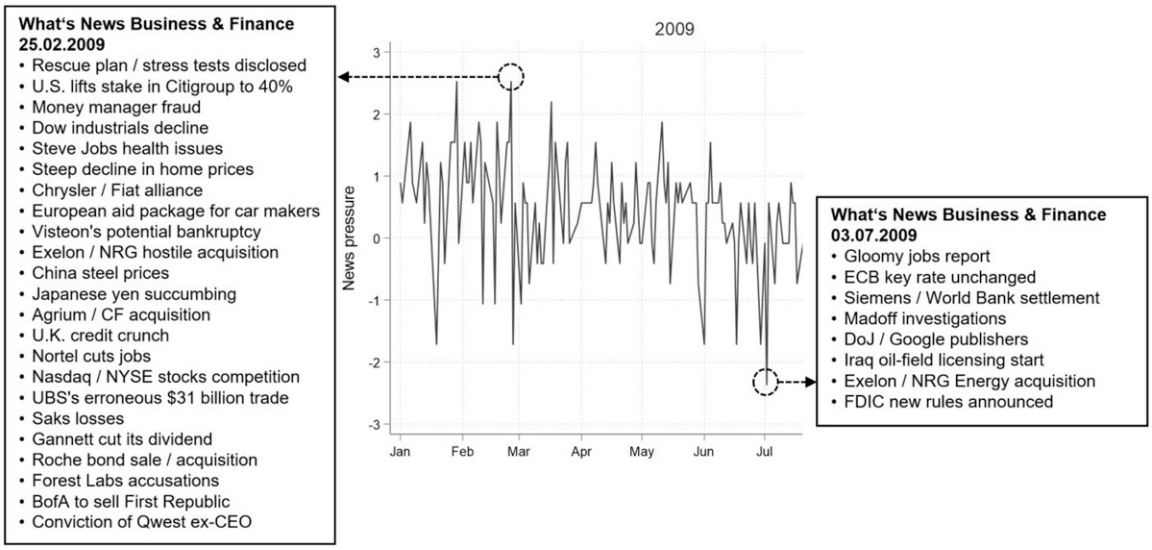
To test Hypothesis 1, our approach is to aggregate the data on the calendar-day level and assess the correlation between news pressure and the number of data breaches announced. However, a simple correlation may hide the fact that news pressure is a function of unpredictable and predictable events. News pressure is substantially driven by unpredictable events (e.g., disasters, scandals, forced chief executive officer (CEO) turnovers) to which data breach announcements could not possibly be timed.

**Figure 1.** News Pressure, 2008–2018




*Notes.* The figure plots the seven-day moving average of (corrected) *NEWS PRESSURE* over the observation period from January 1, 2008, to (and including) December 31, 2018. To account for the layout change of the *WSJ* in August 2013, *NEWS PRESSURE* is standardized within the pre- and postchange periods.

**Figure 2.** Exemplary News Pressure Between January 2009 and July 2009




*Notes*. The figure plots the standardized (corrected) *NEWS PRESSURE* between January 2009 and July 2009, along with the items mentioned in the What's News: Business & Finance column. The item headlines are summarized to preserve space.

To overcome this obstacle, we implement an instrumental variables approach, which aims to isolate the part of the correlation that is due to predictable news pressure. As an instrument, we rely on the count of prescheduled earnings announcements by publicly traded U.S. firms on day $t$.

To be valid, the instrument must fulfill the relevance criterion and the exclusion restriction (Angrist et al. 1996). The relevance criterion states that earnings announcements must be a strong predictor of news pressure, both theoretically and empirically. From a theoretical standpoint, earnings announcements are a relevant determinant of news pressure in the *WSJ* for three reasons. First, as an official public statement of a firm's profitability, earnings announcements represent an important event in the corporate calendar (e.g., Beaver 1968 and deHaan et al. 2015). Business analysts follow these events, which are covered by the trade press, including *WSJ*. For example, the What's News column shown in Section 5 in Online Appendix A reports the earnings announcements of Amazon, Facebook, Intel, BP, and Total. Second, earnings announcements are methodologically suitable because they are prescheduled, which enables breached firms to time their data

**Table 2.** Variables

| Variable | Description |
|---|---|
| log(*BREACH ANNCMTS*) | Count of data breaches announced on $t$, log(var + 1) transformed |
| *BREACH* | Equal to 1 if a data breach was announced on $t$; 0 otherwise |
| log(*BREACHED RECORDS*) | Count of records breached on $t$ (in thousands), log(var + 1) transformed |
| *SEVERE* | Multinomial: 0 = no breach, 1 = no severe breach, 2 = at least one severe breach; severe if an above-median number of records is leaked and if Social Security numbers are leaked |
| *EXTERNAL CAUSE* | Multinomial: 0 = no breach, 1 = firm-external cause [hacking/ransomware], 2 = breach has firm-internal cause [employee error/theft] |
| *PUBLIC FIRM* | Multinomial: 0 = no breach, 1 = private firm, 2 = public firm |
| *DATA TYPE* | Multinomial: 0 = no breach, 1 = financial data 2 = health data, 3 = credentials, 4 = other |
| log(*NEWS PRESSURE*) | Count of articles in the *WSJ* What's News: Business & Finance column associated with $t$, logged |
| *WSJ COVERAGE* | Equal to 1 if What's News reported an article on a data breach on $t$; 0 otherwise |
| log(*EARN ANNCMNTS*) | Count of prescheduled earnings announcements on $t$, log(var + 1) transformed |
| log(*EMPLOYEES*) | Count of employees recorded for the year preceding the announcement, logged |
| log(*EBIT*) | Earnings before interest and taxes for the preceding fiscal year, logged |
| log(*CAPEX*) | Capital expenditures for the preceding fiscal year, logged |
| log(*OPEX*) | Operating expenses for the preceding fiscal year, logged |
| *IT_INDUSTRY* | Equal to 1 if the firm is in the IT sector, as identified by the first two digits of the North American Industry Classification System; 0 otherwise |
| *FIN_INDUSTRY* | Equal to 1 if the firm is in the finance and insurance sector, as identified by the first two digits of the North American Industry Classification System; 0 otherwise |

*Note*. This table defines the variables used in the data sets and states their source.

breach announcement to coincide with other firms' earnings announcements. Finally, it is likely that earnings announcements crowd out other news. In that vein, Hirshleifer et al. (2009) find that investors' same-day reactions to earnings announcements are less sensitive when many firms announce their earnings on the same day. Formal tests reported later in this paper confirm the relevance criterion.

Regarding the exclusion restriction, the instrument must not influence the dependent variable in any way except through the independent variable. There is no formal test for the exclusion restriction; therefore its plausibility remains a theoretical argument. However, several reasons suggest that the exclusion restriction is likely to hold. One reason is temporal exclusion. Earnings announcements are prescheduled and relatively constant within firms. Prior work has documented that managers have a preference for consistency in earnings reporting (Graham et al. 2005). This preference has been attributed to managers' perceptions that markets may react to changes in the timing of an earnings announcement. Analysts make efforts to predict earnings dates, and evidence indicates negative market reactions to firms that delay expected reporting dates (Chambers and Penman 1984). Another reason for the exclusion criterion to hold is external agency. The instrument consists only of earnings announced by firms other than the breach-announcing firm. There is little reason that firms coordinate to time their earnings announcements to data breach announcements because the latter are not prescheduled.

Obtaining correct dates of the earnings announcements is critical for data collection, but the data sources can exhibit deviations (DellaVigna and Pollet 2009, deHaan et al. 2015). Departing from deHaan et al. (2015), the expected accuracy of the dates in our sample is already relatively high (above 90%).[8] Nevertheless, we follow the recommendation and consider only earnings announcements with identical dates in at least two data sources. We obtained quarterly earnings announcements dates for U.S. public firms from the Institutional Brokers' Estimate System (I/B/E/S) for the period 2008–2018 and eliminated those for which I/B/E/S data are unavailable. We then intersected the data with the Compustat Capital IQ Key Developments database and dropped observations for which the dates differed.[9] Because the results would be biased if firms announced data breaches together with their earnings, we also dropped the seven data breaches that were announced within a window of [−1, 1] days of the firms' earnings announcements.[10] The resulting instrumental variable is *EARN ANNCMNTS*, which holds the count of verified earnings announced by U.S. public firms on day *t*. We log-transformed the variable as a result of skewness.

The resulting data set is on the day level and lists all the individual days as units of observation, along with news pressure, the number of announced data breaches, and variables specific to that day. We retain data breaches affecting nonpublic firms in the sample to provide a richer overview on the data but later document that the results hold when restricted to announcements of public firms.[11]

## 5.2. Results

**5.2.1. Test for Hypothesis 1.** To test Hypothesis 1, we estimate a standard instrumental variables model where the first stage is

$$NEWS\ PRESSURE_t = \beta_0 + \beta_1 EARN\ ANNCMNTS_t + \beta_2 X_t + u_t \quad (1)$$

where $t$ indexes for days. The variable list X contains dummies for the day of the week, week, month, and year. The error term is denoted by u. These dummies account for variation that is constant within those time dimensions. In addition, X contains an indicator that equals 1 after the design change of the *WSJ* front page in August 2013. Standard errors are heteroscedasticity robust and, following Durante and Zhuravskaya (2018), clustered for year-month.

Table 3 reports the results. Column (1) presents the first-stage ordinary least squares (OLS), in which news pressure is regressed on the full set of time dummies and the instrument that captures the (logged) number of earnings announcements on that day. This effect is positive and significant. The adjusted $R^2$ of the first stage is 0.726. The Cragg–Donald Wald *F*-test statistic rejects the null hypothesis of a weak instrument. The test statistic is 21.898, which exceeds the critical value of 16.38 proposed by Stock and Yogo (2005). The Kleibergen–Paap rk Wald *F*-statistic is 17.012, which also allays concerns over a weak instrument.[12] Column (2) adds lags and leads for *EARN ANNCMNTS*. The estimate is similar and significant.[13] We interpret the magnitude in terms of standardized effects. Days with a one-standard-deviation-higher number of earnings announcements are associated with 3% higher news pressure.[14]

To test for strategic timing, we fit a two-stage least squares model (2SLS) as follows:

$$BREACH\ ANNCMTS_t = \beta_0 + \beta_1 NEWS\ \widehat{PRESSURE}_t + \beta_2 X_t + u_t \quad (2)$$

where we regress the dependent variable on the predicted (logged) news pressure and the control variable list X, which is identically defined as in equation (1). We estimate the equation with 2SLS. The dependent variable is *BREACH ANNCMTS*.

**Table 3.** Strategic Timing of Data Breach Announcements (Hypothesis 1)

| | First stage | | Second stage | | | | |
|---|---|---|---|---|---|---|---|
| | log(*NEWS PRESSURE*) | | log(*BREACH ANNCMTS*) | | | | log(*BREACHED RECORDS*) |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| log(*EARN ANNCMNTS*)$_t$ | 0.021*** | 0.022*** | | | | | |
| | (0.005) | (0.005) | | | | | |
| log(*EARN ANNCMNTS*)$_{t\text{-}1}$ | | −0.010* | | | | | |
| | | (0.004) | | | | | |
| log(*EARN ANNCMNTS*)$_{t+1}$ | | −0.000 | | | | | |
| | | (0.005) | | | | | |
| log(*NEWS PRESSURE*)$_t$ | | | 0.134* | 4.361** | 3.918*** | 3.869*** | 19.916* |
| | | | (0.068) | (1.368) | (1.154) | (1.130) | (7.859) |
| log(*NEWS PRESSURE*)$_{t\text{-}1}$ | | | | | −0.569** | −0.556** | −3.214** |
| | | | | | (0.187) | (0.182) | (1.159) |
| log(*NEWS PRESSURE*)$_{t+1}$ | | | | | −0.600** | −0.575** | −3.834** |
| | | | | | (0.218) | (0.211) | (1.428) |
| log(*BREACH ANNCMTS*)$_{t\text{-}1}$ | | | | | | 0.010 | −0.215 |
| | | | | | | (0.029) | (0.224) |
| log(*BREACH ANNCMTS*)$_{[7\text{-day average}]}$ | | | | | | −0.146$^+$ | −1.237$^+$ |
| | | | | | | (0.078) | (0.699) |
| log(*BREACH ANNCMTS*)$_{[14\text{-day average}]}$ | | | | | | 0.340* | 1.163 |
| | | | | | | (0.135) | (1.067) |
| Cragg–Donald Wald *F* | 21.898 | 22.724 | | | | | |
| Kleibergen–Paap rk Wald *F* | 17.012 | 17.886 | | | | | |
| Time fixed effects | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Estimator | OLS | OLS | OLS | 2SLS | 2SLS | 2SLS | 2SLS |
| *N* | 2,757 | 2,755 | 2,757 | 2,757 | 2,755 | 2,742 | 2,742 |
| Adjusted $R^2$ | 0.726 | 0.726 | 0.250 | — | — | — | — |

*Notes.* Estimations are based on the calendar-day data set. Standard errors are in parentheses. Time fixed effects are dummies for year, month, week, and day of the week. The number of observations differs because of the inclusion of lags and leads.

$^+$, *, **, and *** indicate significance at the 10%, 5%, 1%, and 0.1% levels, respectively.

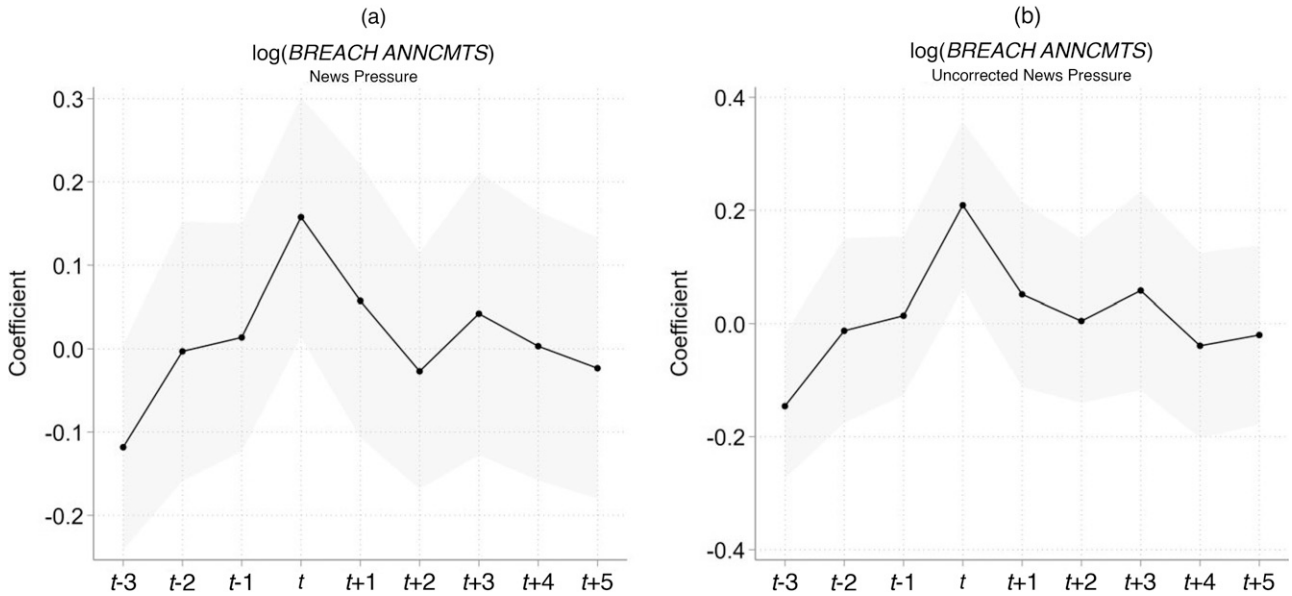
Before estimating the full 2SLS model, we explore the simple correlation, reported in column (3). We observe a significant positive correlation between news pressure and data breach announcements. To estimate the effect sizes, we follow Lin et al. (2013). A 1% higher level of news pressure is associated with $100(e^{0.134\times\ln(1.01)}-1) = 0.13\%$ more data breach announcements. To further illustrate the correlation, Figure 3 plots the coefficients on the lags and leads of news pressure for [−3, 5] days around the announcement together with 95% confidence intervals. News pressure on day $t$ is positively and significantly correlated with the number of data breach announcements.[15]

Continuing with Table 3, column (4) reports the main evidence for strategic timing: a positive correlation between predictable news pressure and the number of data breach announcements. The effect is positive and significant. A 1% higher level of news pressure is associated with $100(e^{4.361\times\ln(1.01)} - 1) = 4.44\%$ more data breaches announced that day. Thus, firms are more likely to announce data breaches on days with predictably high news pressure.[16] Columns (5)–(7) further corroborate our argument for strategic timing. Column (5) adds a one-day lag and lead in news pressure. The estimated effect decreases marginally to $100(e^{3.918\times\ln(1.01)} - 1) = 3.98\%$. Column (6) addresses potential concerns over serial correlation in the data, in terms of the number of data breach announcements on day $t$ correlating with the number of data breach announcements on preceding or subsequent days. We add the average number of data breach announcements made over the 7 and 14 days before the announcement, as well as a lag. The results are consistent. At $100(e^{3.869\times\ln(1.01)} - 1) = 3.93\%$, the estimated effect is marginally smaller. Column (7) estimates *BREACHED RECORDS* as an alternative dependent variable. A one-standard-deviation-higher level of predictable news pressure is associated with approximately 19.024 more individual records announced to be breached.[17]

Table 4 replicates the analysis for uncorrected news pressure. The estimates are consistent in magnitude and significance with the estimates for the corrected news pressure, which further corroborates the conclusion that data breach announcements coincide with days that show higher levels of predictable news pressure. Taken together, these results support Hypothesis 1.

**5.2.2. Heterogeneity.** Next, we examine heterogeneity in the estimates to explore whether firms' incentives for strategic timing differ depending on breach or

**Figure 3.** Data Breach Announcements and News Pressure




*Notes.* The figure reports the estimated coefficients with 95% confidence intervals from the regressions of the count of data breach announcements on news pressure between three days before and five days after. Panel A uses the corrected news pressure measure; panel B uses the uncorrected news pressure.

firm characteristics. Econometrically, we create multinomial response variables and fit a multinomial logistic regression model.[18]

Table 5 summarizes the analyses. First, we examine whether the severity of a data breach renders strategic timing more likely (e.g., Malhotra and Malhotra 2011). If a breach is severe, firms are likely to face greater consequences and should therefore have greater incentives to avoid media attention. To that end, we regress news pressure on *SEVERE*. In panel A, we observe that the correlation is stronger for severe breaches (Wald $\chi^2 = 23.96$, $p < 0.001$). Section 2 in Online Appendix B reports an extension of this test wherein the response variable gives the deciles of *BREACHED RECORDS*. We find that strategic timing is consistently present for breaches above the 60th percentile.[19]

Second, we examine whether firm-external and firm-internal causes of data breaches induce heterogeneity. External causes, such as hacking, may allow a firm to be portrayed as a victim in the media, whereas internal causes, such as employee errors, make it likely that a firm will be framed as a perpetrator and reprimanded for poor security practices (see Weaver 2007). Thus, we expect that firms will be more likely to engage in strategic timing when data breaches are caused internally. We test this prediction by regressing news pressure on *EXTERNAL CAUSE*. Confirming our expectation, panel B shows that the correlation is stronger for data breach announcements based on firm-internal causes than for those based on firm-external causes (Wald $\chi^2 = 26.05$, $p < 0.001$).

Third, we examine heterogeneity depending on whether a data breach is announced by public as opposed to private firms (*PUBLIC FIRM*). Given that the markets and the media pay more attention to public firms, we expect public firms to rely more heavily on strategic timing. In panel B, we find that this is indeed the case. The correlation is stronger for data breaches announced by public firms than for those announced by private firms (Wald $\chi^2 = 25.43$, $p < 0.001$).

Fourth, we explore heterogeneity in terms of the type of data breached (*DATA TYPE*). We distinguish between financial data (i.e., coded in ITRC as "credit card" and "bank accounts"), health data (i.e., coded as such by ITRC), credentials (i.e., coded as "Emails" and "Username/Password"), and other (i.e., all remaining classifications, including "DMV [Department of Motor Vehicle] records" and "Other"). As reported in panel C, we find the strongest correlation between news pressure and data breach announcements for those that exposed health data, financial data, and credentials (joint Wald $\chi^2 = 24.95$, $p < 0.001$).

### 5.3. Robustness

We conduct the following robustness checks. First, we complement our day-level estimation with a firm-level analysis, which is reported in Online Appendix B, Section 1. We estimate a survival model where firms are metaphorically at risk for making a data breach announcement after discovering a data breach.

**Table 4.** Strategic Timing of Data Breach Announcements (Uncorrected News Pressure)

| | First stage | | Second stage | | | | |
|---|---|---|---|---|---|---|---|
| | log(*NEWS PRESSURE UNC.*) | | log(*BREACH ANNCMTS*) | | | | log(*BREACHED RECORDS*) |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| log(*EARN ANNCMNTS*)$_t$ | 0.020*** | 0.021*** | | | | | |
| | (0.005) | (0.005) | | | | | |
| log(*EARN ANNCMNTS*)$_{t\text{-}1}$ | | −0.011* | | | | | |
| | | (0.004) | | | | | |
| log(*EARN ANNCMNTS*)$_{t+1}$ | | −0.001 | | | | | |
| | | (0.005) | | | | | |
| log(*NEWS PRESSURE UNCORRECTED*)$_t$ | | | 0.180* | 4.489** | 3.994*** | 3.939*** | 20.118* |
| | | | (0.069) | (1.417) | (1.172) | (1.147) | (7.920) |
| log(*NEWS PRESSURE UNCORRECTED*)$_{t\text{-}1}$ | | | | | −0.584** | −0.573** | −3.431** |
| | | | | | (0.196) | (0.191) | (1.193) |
| log(*NEWS PRESSURE UNCORRECTED*)$_{t+1}$ | | | | | −0.628** | −0.603** | −3.905** |
| | | | | | (0.227) | (0.220) | (1.473) |
| log(*BREACH ANNCMTS*)$_{t\text{-}1}$ | | | | | | 0.014 | −0.191 |
| | | | | | | (0.029) | (0.223) |
| log(*BREACH ANNCMTS*)$_{[7\text{-day average}]}$ | | | | | | −0.138$^+$ | −1.200$^+$ |
| | | | | | | (0.077) | (0.688) |
| log(*BREACH ANNCMTS*)$_{[14\text{-day average}]}$ | | | | | | 0.312* | 1.035 |
| | | | | | | (0.135) | (1.056) |
| Cragg–Donald Wald *F* | 21.709 | 22.946 | | | | | |
| Kleibergen–Paap rk Wald *F* | 16.396 | 17.612 | | | | | |
| Time fixed effects | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Estimator | OLS | OLS | OLS | 2SLS | 2SLS | 2SLS | 2SLS |
| *N* | 2,757 | 2,755 | 2,757 | 2,757 | 2,755 | 2,742 | 2,742 |
| Adjusted $R^2$ | 0.730 | 0.730 | 0.251 | — | — | — | — |

*Notes*. Estimations are based on the calendar-day data set. Standard errors are in parentheses. Time fixed effects are dummies for year, month, week, and day of the week. The number of observations differs because of the inclusion of lags and leads.

$^+$, *, **, and *** indicate significance at the 10%, 5%, 1%, and 0.1% levels, respectively.

**Table 5.** Heterogeneity in Strategic Timing by Firm and Data Breach Characteristics

| | Panel A: By breach severity | | | |
|---|---|---|---|---|
| | *SEVERE* | | | |
| | Not severe (=1) | | Severe (=2) | |
| log(*NEWS PRESSURE*) | 19.954*** | | 20.980*** | |
| | (4.169) | | (5.626) | |
| Wald test ($\chi^2$) | 23.96*** | | | |
| | **Panel B: By cause and firm type** | | | |
| | *EXTERNAL CAUSE* | | *PUBLIC FIRM* | |
| | External (=1) | Internal (=2) | Private (=1) | Public (=2) |
| log(*NEWS PRESSURE*) | 10.078* | 26.130*** | 18.893*** | 24.074*** |
| | (4.081) | (5.128) | (4.216) | (5.173) |
| Wald test ($\chi^2$) | 26.05*** | | 25.43*** | |
| | **Panel C: By type of data breached** | | | |
| | *DATA TYPE* | | | |
| | Financial (=1) | Health (=2) | Credentials (=3) | Other (=4) |
| log(*NEWS PRESSURE*) | 22.639*** | 21.895*** | 22.007** | 17.935*** |
| | (5.491) | (5.249) | (6.786) | (4.787) |
| Joint Wald test ($\chi^2$) | 24.95*** | | | |

*Notes*. Estimations are based on the calendar-day data set. $N$ = 2,757 days. Multinomial logit estimates. For brevity, the pairwise Wald tests for panel C are reported in the online appendix. Standard errors are in parentheses.

*, **, and *** indicate significance at the 5%, 1%, and 0.1% levels, respectively.

We find that higher expected news pressure significantly reduces the time to announcement. This analysis is limited by the fact that we can observe the discovery dates for a fraction of the firms in the sample. In a second analysis, we find the correlation confirmed on the firm level for different windows around the actual announcement. Second, we replicate the analysis when data breach announcements are restricted to only those of public firms. Table 6 reports the second-stage results when restricting data breach announcements to those made by public firms. The results are consistent.

Third, we confirm that data breach announcements are covered by the *WSJ* and that news pressure is associated with less coverage of data breach announcements, two main premises of our research design. In Table 7, column (1) shows that the odds of the *WSJ* covering a data breach are larger the more data breaches are recorded for that day. Columns (2) and (3) reveal that the odds of coverage are also higher when public firms are breached or when data breaches are severe.[20] In column (4), we observe that higher news pressure indeed crowds out *WSJ* reporting on data breach announcements. Taken together, these analyses—although correlational—support the main premises of our research.

Fourth, we test whether our results are robust to alternative operationalizations of news pressure. Column (5) in Table 6 reports the results for not log-transformed news pressure. Columns (6) and (7) report the results when operationalizing news pressure based on the logged character and word count of the *WSJ* column, respectively. Our findings are robust to all these alternatives.

Fifth, we test whether our results are robust to alternative operationalizations of data breach announcements and sample selection bias (see Online Appendix B, Section 3). We tested a binary version of the variable, a not log-transformed version, a subsample of data breach announcements based on official filings with authorities, and a subsample of data breach announcements excluding Fridays. The results are robust against all these alternative operationalizations and subsamples.

Sixth, we assess the sensitivity of the instrument by replicating the analysis with the instrument weighted by the market value of the firms that report their earnings (see Online Appendix B, Section 4). Again, the results are consistent.

Seventh, we test for rival explanations (see Online Appendix B, Section 5). Here, we find that Fridays are associated with a significantly higher number of data breach announcements. This finding is further corroborating our hypothesis that firms might time the announcement to avoid attention. DellaVigna and Pollet (2009) argued that Fridays may constitute a period of market inattention because of the upcoming weekend. In addition, we find that strategic timing is less prevalent for announcements that the ITRC data set recorded from newswire services (e.g., PR Newswire). The use of newswire agencies might weaken the impact of strategic timing because such agencies have fewer incentives to time an announcement and because the announcement is likely to reach a larger and wider audience.[21]

Eighth, we conduct a placebo test using an unpredictable source of news pressure as an instrument to which firms cannot time their actions. A significant first stage but insignificant second stage here would indirectly support our results. In summary, using the

**Table 6.** Second-Stage Estimates for Announcements of Public Firms

| | log(*BREACH ANNCMTS PUBLIC FIRMS*) | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| log(*NEWS PRESSURE*)$_t$ | 0.072$^+$ | 1.291* | 1.265* | 1.252* |
| | (0.043) | (0.594) | (0.525) | (0.515) |
| log(*NEWS PRESSURE*)$_{t\text{-}1}$ | | | −0.151$^+$ | −0.150* |
| | | | (0.077) | (0.076) |
| log(*NEWS PRESSURE*)$_{t+1}$ | | | −0.214* | −0.208* |
| | | | (0.099) | (0.096) |
| log(*BREACH ANNCMTS*)$_{t\text{-}1}$ | | | | −0.017 |
| | | | | (0.047) |
| log(*BREACH ANNCMTS*)$_{[7\text{-day average}]}$ | | | | 0.055 |
| | | | | (0.063) |
| log(*BREACH ANNCMTS*)$_{[14\text{-day average}]}$ | | | | 0.014 |
| | | | | (0.021) |
| Time fixed effects | Yes | Yes | Yes | Yes |
| Estimator | OLS | 2SLS | 2SLS | 2SLS |
| *N* | 2,757 | 2,757 | 2,755 | 2,742 |

*Notes.* Estimations are based on the calendar-day data set. Standard errors are in parentheses. Time fixed effects are dummies for year, month, week, and day of the week. The number of observations differs because of the inclusion of lags and leads.

$^+$ and * indicate significance at the 10% and 5% levels, respectively.

**Table 7.** Robustness Checks for Assumptions of Study 1 and News Pressure Measure

| | Test for assumptions | | | | Alternative news pressure measures | | |
|---|---|---|---|---|---|---|---|
| | *WSJ COVERAGE* | | | | log(*BREACH ANNCMTS*) | | |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| log(*BREACH ANNCMTS*) | 0.365* | | | | | | |
| | (0.148) | | | | | | |
| *PUBLIC FIRM* | | 0.380* | | | | | |
| | | (0.182) | | | | | |
| *SEVERE* | | | | | | | |
| Not severe (=1) | | | 0.117 | | | | |
| | | | (0.242) | | | | |
| Severe (=2) | | | 0.798** | | | | |
| | | | (0.278) | | | | |
| log(*NEWS PRESSURE*) | | | | −5.857*** | | | |
| | | | | (0.657) | | | |
| *NEWS PRESSURE* | | | | | 0.381** | | |
| | | | | | (0.126) | | |
| log(*CHARACTER COUNT*) | | | | | | 7.542* | |
| | | | | | | (3.480) | |
| log(*WORD COUNT*) | | | | | | | 8.403+ |
| | | | | | | | (4.297) |
| Time fixed effects | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Estimator | Logit | Logit | Logit | Logit | 2SLS | 2SLS | 2SLS |

*Notes*. Estimations are based on the calendar-day data set. $N = 2{,}757$ days. Standard errors are in parentheses.
+, *, **, and *** indicate significance at the 10%, 5%, 1%, and 0.1% levels, respectively.

announcement of forced CEO turnovers by U.S. public firms as an unpredictable source of news pressure, we find that, in the first stage, they are related to higher levels of news pressure, yet the second stage is, as expected, insignificant. This analysis corroborates our research design. Section 6 in Online Appendix B provides the motivation, details on the data collection, and the results for this analysis.

Finally, given that the breach announcements in the sample are subject to different notification requirements, it is important to assess how this heterogeneity affects the results.[22] Ideally, such an analysis would require a state-level investigation akin to a policy evaluation; however, this option is not available to us. The best we can do for this analysis is to proceed as we did in the heterogeneity analyses and create a multinomial variable that equals 0 on days with no announcements, 1 on days with announcements that were not subject to a fixed notification deadline (i.e., "most expeditious time possible"), and 2 on days with announcements that were subject to a fixed notification deadline (e.g., "maximum of 30 days"). We find that strategic timing is more prevalent among announcements that were not subject to a notification deadline. However, the magnitude of the difference is small—perhaps because even fixed notification deadlines continue to allow substantial degrees of freedom in the timing of firms' announcements. Thus, although the notification requirements differ within the sample, we do not observe that this harms the interpretation of the findings. Section 7 in Online Appendix B provides the details on the data coding and the results.[23]

## 6. Study 2: News Pressure and Market Reactions (Hypothesis 2)

### 6.1. Research Design

To test Hypothesis 2, we follow prior research on data breaches by implementing an event study at the firm level (e.g., Goel and Shawky 2009). An event study assumes that the market impact of an event can be observed via abnormal changes in a firm's stock price (e.g., Brown and Warner 1985). An abnormal return is the difference between the observed return and the return expected if the event would not have occurred. In other words, we estimate a counterfactual return for each firm and compare it to the firm's actual return. We build on this notion and argue that a firm announcing a data breach incurs negative performance impacts, such as a loss of customers, a loss of reputation, and entanglement in litigation, which should become reflected in the firm's stock price.[24]

To conduct the event study, we transformed the data set into a firm-level cross section. We restricted the sample to public firms and matched each firm's name to its permanent company identifier (LPERMCO) in the Center for Research in Security Prices (CRSP) daily database. To avoid multiple treatments of the same firm involved in multiple breaches, we kept only each firm's first breach in

the sample. To reduce the possibility of confounding events, we merged the identified events with the Capital IQ Key Developments database and excluded events that involved merger and acquisition activity and board changes in the same week. Section 1 in Online Appendix C reports further details on the sample construction, including the treatment of multiple breaches, confounding events, and further cleansing.

The final sample comprises 311 firms. The loss in observations compared with the full ITRC data set is mostly caused by the lack of pre-event CRSP data and because data are not sufficient for estimating the expected returns for some firms. Still, this sample is relatively large compared with other studies (cf. Online Appendix A, Section 2).[25]

### 6.2. Econometric Framework

To estimate market reactions, we follow the standard financial economics literature and estimate abnormal returns (ARs) as follows:

$$AR_{i,t} = r_{i,t} - \hat{r}_{i,t}, \tag{3}$$

where $r_{i,t}$ is the actual stock return of firm $i$ on trading day $t$ and $\hat{r}_{i,t}$ is the expected return. The expected returns are predicted for a pre-event window and are based on different models. Three expected returns models are commonly used in current event studies: the market model (MM), the Fama–French three-factor (FF3) model, and the Fama–French–Carhart four-factor (FFC4) model. These models all use the market returns to estimate $\hat{r}_{i,t}$ but differ in their inclusion of additional covariates.

We follow extant event studies of data breach announcements and rely primarily on a market model while documenting robustness toward alternative approaches (Sorescu et al. 2017). Based on the MM, the expected returns are as follows:

$$\hat{r}_{i,t} = \alpha + \beta_i(r_{m,t} - r_{f,t}) + \epsilon_{i,t}, \tag{4}$$

where $r_{m,t}$ is the average rate of return for all stocks trading in market index $m$ wherein firm $i$ is listed, and $r_{f,t}$ equals the risk-free rate of return at time $t$.

Abnormal returns are cumulated (CARs) over an event window as follows:

$$CAR(t_1, t_2)_i = \sum_{t=t_1}^{t_2} AR_{i,t}, \tag{5}$$

where $t_1$, $t_2$ are the lower and upper boundaries of the event window, respectively. The parameters are as follows. First, we use an event window of [−3, 5] to allow for the leakage of information and to allow the markets to impound the effects into stock prices. Whereas longer event windows allow the gradual diffusion of information, they must be weighed against a corresponding higher risk of contamination from extraneous events (e.g., Sorescu et al. 2017). To assess sensitivity, we report the estimates from various windows (i.e., [0], [0, 1], [−1, 1], [−3, 10]). Second, we rely on a pre-event period of [−100, −30]. The returns are adjusted for nontrading days such that nontrading dates are converted to the next trading day. We show robustness toward a longer estimation window.

To test for Hypothesis 2, our estimation model then is as follows:

$$CAR_i = \beta_0 + \beta_1 NEWS\ PRESSURE_t + \beta_2 X_t + \beta_3 V_i + u_i, \tag{6}$$

where *CAR* is defined for firm $i$, and *NEWS PRESSURE* is defined for the announcement day $t$; $X$ contains time fixed effects, and $V$ contains firm-specific controls. The intercept $\beta_0$ is the estimate of the CARs when news pressure is 0 (i.e., for this purpose, we center *NEWS PRESSURE*), and $\beta_1$ adjusts the estimate for different levels of news pressure. Standard errors are clustered for year-quarter, given limited variation within year-months.

We use five firm-specific controls identified from the Compustat Capital IQ database. First, we proxy for firm size by *EMPLOYEES*, which indicates the (logged) number of employees recorded for the year preceding the announcement (e.g., Kohli et al. 2012). The direction of the influence of firm size is difficult to predict. Larger firms may be more susceptible to data breaches, yet markets may also consider large firms to be more capable of overcoming the negative outcomes of such breaches. Second, we control for a firm's earnings structure by including its (logged) earnings before interest and taxes (*EBIT*) for the preceding fiscal year (e.g., Kohli et al. 2012). Third, we control for the investments a firm makes via its (logged) capital expenditures (*CAPEX*). *CAPEX* includes the funds a firm uses for additional property, plants, and equipment but excluding those arising from acquisitions. Higher investments may produce higher expectations of future returns, or they may indicate investments into IT. Fourth, we control for more general business costs by including a firm's (logged) operating expenses (*OPEX*). *OPEX* reflects the cost of doing business and enable us to adjust for the different types of business models employed by firms in the sample and for their use of third parties in delivering IT infrastructure (e.g., Mani et al. 2013). Finally, we control for industry. The dummy *IT_INDUSTRY* equals 1 if the firm is associated with the IT industry based on the first two digits of the North American Industry Classification System (e.g., Han and Mithas 2013). Given the reduced sample size, we include only day-of-week and year dummies.

## 6.3. Results

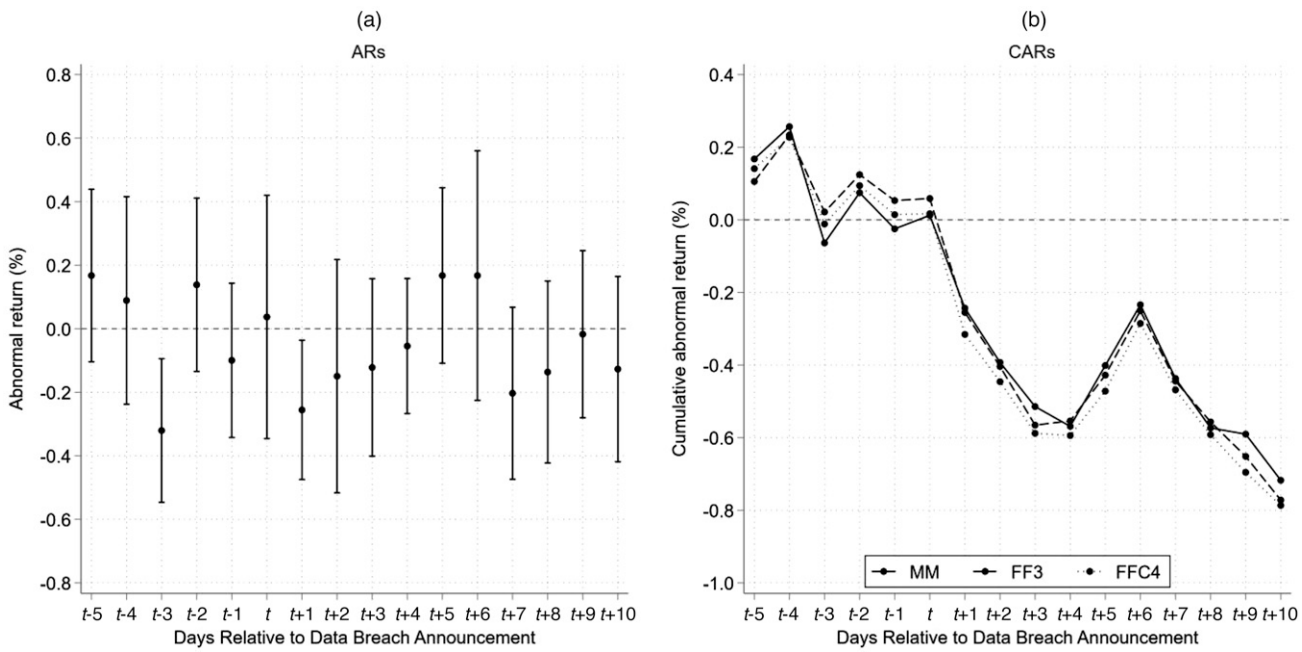
**6.3.1. Market Reactions Without Accounting for News Pressure.** Before testing Hypothesis 2, we begin by assessing the market reactions without accounting for news pressure. Figure 4, panel A plots the ARs together with 95% confidence intervals. We observe that several ARs over the event window are negative, with the ones on $t - 3$ and $t + 1$ being statistically significant. To ease interpretation, we follow extant event studies (e.g., Acquisti et al. 2006 and Telang and Wattal 2007) and plot the CARs over 5 days before the event to 10 days after the event in panel B. There, the solid line represents the MM, the dashed line the FF3 model, and the dotted line the FFC4 model. On the days leading up to the announcement, the CARs are close to 0. Beginning with the day after the announcement, the CARs turn negative. Aside from a small peak at $t + 6$, the CARs decline to below −0.60%.

Table 8 reports formal significance tests for the CARs over different event windows. Following prior event studies of data breach announcements, we use three test statistics. The first test statistic (i.e., the first asterisk) is a one-tailed $t$-test (e.g., Martin et al. 2017). The second test statistic is a signed rank test (e.g., Telang and Wattal 2007). The third test statistic, proposed by Brown and Warner (1985), accounts for event time clustering (e.g., Telang and Wattal 2007). The columns report the CARs for different event windows. The CARs are consistently negative across all specifications, and their magnitude increases with the length of the event window. The CARs are consistently significant for [−3, 5] and [−3, 10], with the only exception being the CARs estimated from the FFC4 model for [−3, 5].

**6.3.2. Test for Hypothesis 2.** Figure 5 presents nonparametric evidence for Hypothesis 2. It plots the CARs for the main event window against equal-sized bins of (standardized) corrected news pressure (panel A) and uncorrected news pressure (panel B), together with a fitted line. The plots suggest a clear relationship between news pressure and CARs: higher values of news pressure are associated with less negative CARs.[26]

Table 9 reports the formal test for Hypothesis 2. We observe evidence in line with the nonparametric evidence: higher levels of news pressure on the announcement day attenuate negative market reactions to data breach announcements. Column (1), which is the baseline, reports the market reactions for an event window of [−3, 5], adjusted for news pressure on the announcement day and for time fixed effects. Column (2) adds controls. The coefficient on *NEWS PRESSURE* is positive and statistically significant, suggesting that a higher level of news pressure on the day of the announcement is associated with less negative market reactions. The intercept is negative and significant, which reflects the negative market reactions to

**Figure 4.** Abnormal and Cumulative Abnormal Returns (in Percent) to Data Breach Announcements




*Notes.* Panel A shows the day-specific abnormal returns with 95% confidence intervals. Panel B cumulates the abnormal returns from 5 days before the data breach announcement to 10 days afterward. Expected returns are derived from MM (solid line), FF3 (dashed line), and FFC4 (dotted line).

**Table 8.** Cumulative Abnormal Returns (in Percent) to Data Breach Announcements

| Model | (1) CAR [−3, 5] | (2) CAR [0, 1] | (3) CAR [−1, 1] | (4) CAR [−1, 3] | (5) CAR [−3, 10] |
|---|---|---|---|---|---|
| MM | $-0.456^{+/+/+}$ | $-0.143^{ns/ns/ns}$ | $-0.241^{ns/+/ns}$ | $-0.357^{ns/+/ns}$ | $-0.791^{+/*/**}$ |
| FF3 | $-0.501^{+/*/+}$ | $-0.249^{ns/+/+}$ | $-0.335^{+/*/ns}$ | $-0.467^{+/+/+}$ | $-0.811^{*/*/*}$ |
| FFC4 | $-0.480^{+/*/ns}$ | $-0.230^{ns/+/ns}$ | $-0.309^{+/+/ns}$ | $-0.480^{*/+/ns}$ | $-0.808^{*/*/*}$ |

*Notes.* $N = 311$ firms. The first test statistic is a one-tailed $t$-test; the second is a signed rank test; and the third, proposed by Brown and Warner (1985), accounts for event time clustering. MM, market model; FF3, Fama–French three-factor model; FFC4, Fama–French–Carhart four-factor model.

$^{ns}$Not significant.

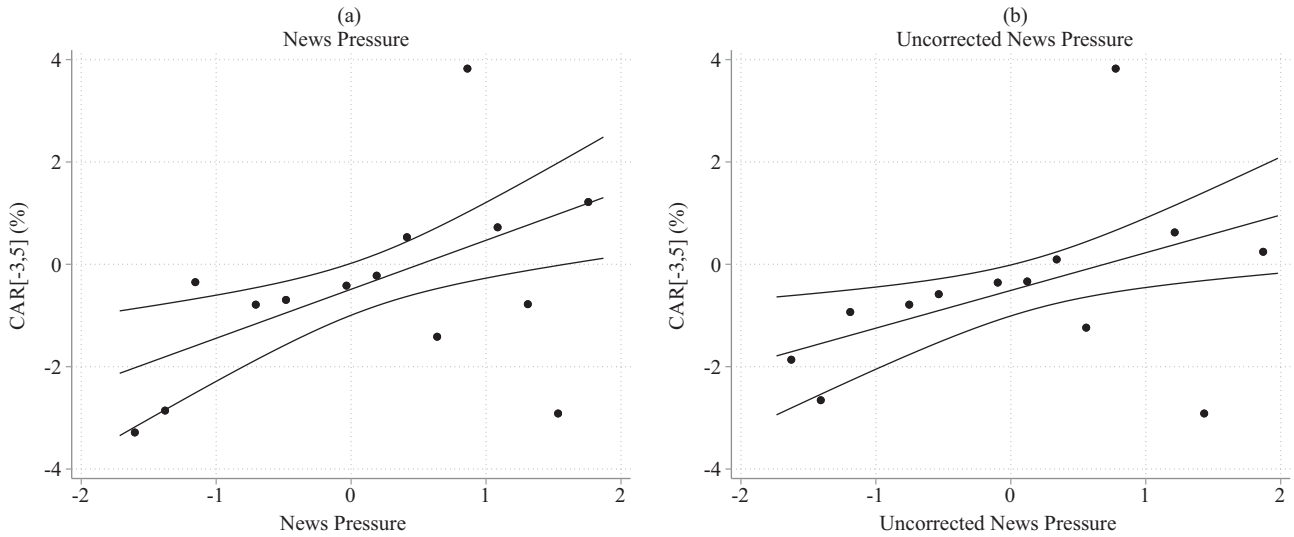
$^{+}$, *, and ** indicate significance at the 10%, 5%, and 1% levels, respectively.

data breach announcements. The remainder of the columns proceed identical but for different sizes of the event window. Across the models, the coefficient on *NEWS PRESSURE* is consistently positive, and the intercept is consistently negative. Table 10 reports the results for uncorrected news pressure. Consistent to before, for the event window, the coefficient on *NEWS PRESSURE* is positive and significant; the intercept is negative and significant. Overall, the estimates are consistent with the ones for corrected news pressure. Taken together, this evidence supports Hypothesis 2.

Figure 6 plots the predicted CARs for different levels of corrected and uncorrected *NEWS PRESSURE* to ease the interpretation of the effect magnitudes. Panel A shows the predicted CARs from column (1) (i.e., the baseline), and panel B plots the predicted CARs from column (2) of Table 9 (i.e., with controls). Panels C and D display the corresponding predicted CARs for uncorrected news pressure. We consistently observe that higher levels of news pressure on the announcement day attenuate the negative market reactions.

We also estimate the market capitalization effects. We infer the market capitalization effects from the product of CAR, the close price of the share before the event window, and the number of outstanding shares (e.g., Bose and Leung 2019). From the event window [−3, 5], we estimate that if news pressure is on its empirical mean, a data breach announcement is associated with a median (respectively, average) decline in market capitalization of $347 million (respectively, $2,257 million). If news pressure is one standard deviation above the mean, we estimate that the loss of market capitalization is less marked, amounting to $85 million (respectively, $560 million). If news pressure is one standard deviation below the mean, we estimate a market capitalization loss of $616 million (respectively, $4,013 million).

**Figure 5.** Cumulative Abnormal Returns (in Percent) and News Pressure




*Notes.* CARs for the event window are plotted against corrected and uncorrected news pressure. Dots are bins of data breach announcements. We employ the default bin sizing in the cmogram function in Stata. The line is fitted together with 95% confidence intervals.

**Table 9.** Cumulative Abnormal Returns (in Percent) to Data Breach Announcements Adjusted for News Pressure (Hypothesis 2)

| | (1) CAR [−3, 5] | (2) CAR [−3, 5] | (3) CAR [0, 1] | (4) CAR [0, 1] | (5) CAR [−1, 1] | (6) CAR [−1, 1] | (7) CAR [−1, 3] | (8) CAR [−1, 3] | (9) CAR [−3, 10] | (10) CAR [−3, 10] |
|---|---|---|---|---|---|---|---|---|---|---|
| log(*NEWS PRESSURE*) | 7.892*** | 7.136* | 3.883* | 4.643$^{+}$ | 5.415** | 5.077* | 5.432* | 4.588 | 7.300** | 7.960* |
| | (1.926) | (2.948) | (1.625) | (2.583) | (1.789) | (2.422) | (2.143) | (2.848) | (2.498) | (3.695) |
| log(*EBIT*) | | −0.300 | | −0.186 | | −0.156 | | −0.118 | | 0.440 |
| | | (0.435) | | (0.211) | | (0.215) | | (0.319) | | (0.581) |
| log(*EMPLOYEES*) | | −0.229 | | 0.068 | | −0.158 | | −0.279 | | −0.506 |
| | | (0.337) | | (0.166) | | (0.177) | | (0.224) | | (0.570) |
| log(*CAPEX*) | | −0.590 | | 0.027 | | −0.112 | | −0.423 | | −0.706 |
| | | (0.498) | | (0.184) | | (0.243) | | (0.309) | | (0.582) |
| log(*OPEX*) | | 1.313 | | −0.009 | | 0.416 | | 0.892 | | 0.953 |
| | | (0.960) | | (0.316) | | (0.288) | | (0.608) | | (1.350) |
| *IT_INDUSTRY* | | 0.317 | | −0.947 | | −1.104$^{+}$ | | −0.468 | | 0.716 |
| | | (0.875) | | (0.639) | | (0.628) | | (0.700) | | (1.239) |
| Intercept ($\beta_0$) | −4.972*** | −9.078$^{+}$ | −2.294* | −1.221 | −3.409*** | −4.167** | −3.309** | −5.341 | −5.467** | −10.924 |
| | (0.879) | (4.673) | (0.900) | (1.481) | (0.836) | (1.420) | (0.923) | (3.241) | (1.628) | (7.106) |
| Time fixed effects | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Model | MM | MM | MM | MM | MM | MM | MM | MM | MM | MM |
| $N$ | 311 | 208 | 311 | 208 | 311 | 208 | 311 | 208 | 311 | 208 |
| $R^2$ | 0.048 | 0.083 | 0.062 | 0.126 | 0.071 | 0.146 | 0.051 | 0.091 | 0.025 | 0.059 |

*Notes.* OLS estimates; $N$ is in firms. MM, market model. Time fixed effects are dummies for the year and day of the week. Standard errors are in parentheses.

$^{+}$, *, **, and *** indicate significance at the 10%, 5%, 1%, and 0.1% levels, respectively.

## 6.4. Robustness

We implement several robustness checks. First, crucial is the estimation of the counterfactual returns $\hat{r}_{i,t}$ (i.e., the returns had the event not taken place), similar to the validity of the control group in quasiexperimental designs (e.g., Foerderer et al. 2018). Our primary approach is a market model, but the FF3 and FFC4 models offer alternative theoretical conceptualizations of

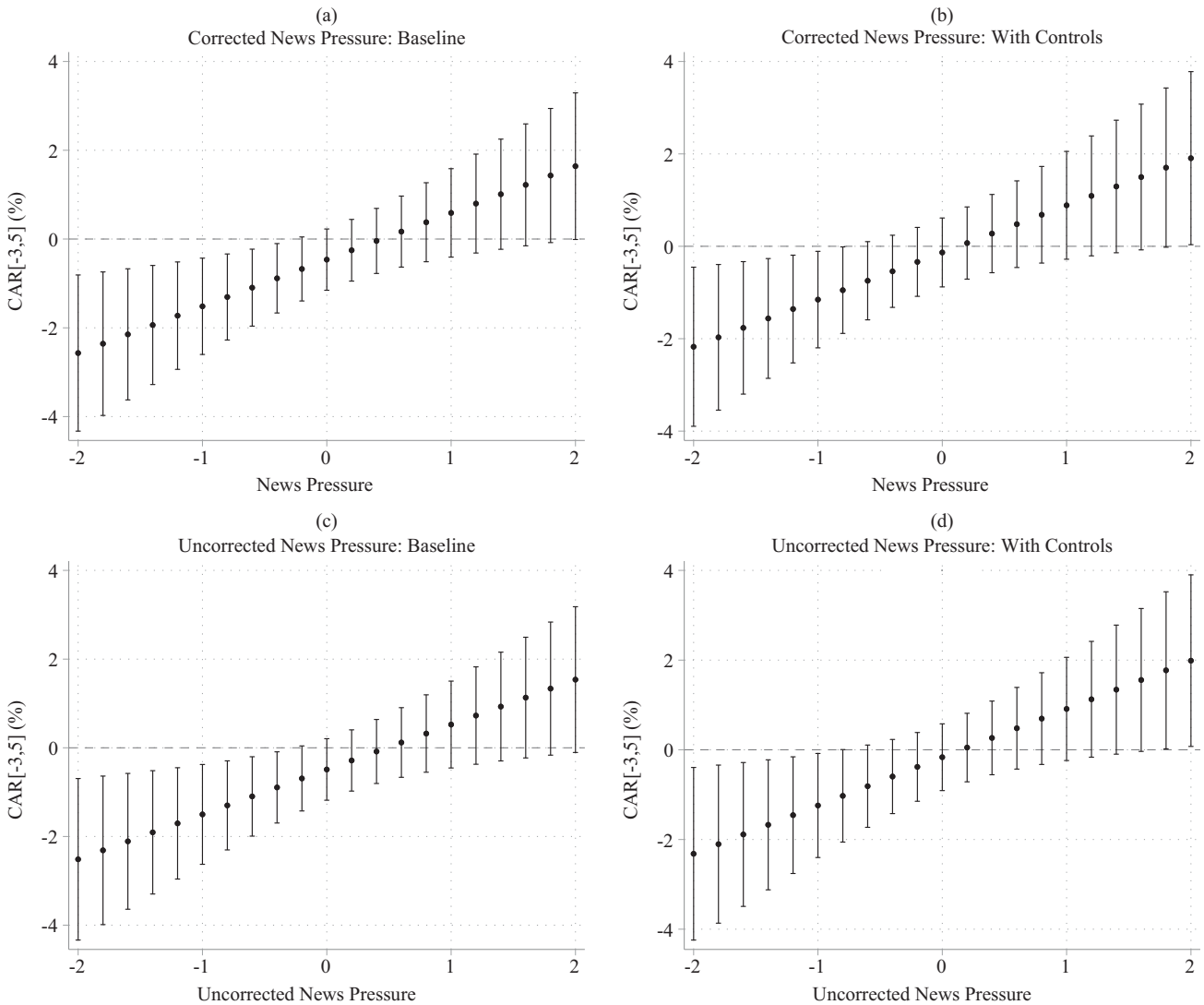
**Table 10.** Cumulative Abnormal Returns (in Percent) to Data Breach Announcements Adjusted for Uncorrected News Pressure

| | (1) CAR [−3, 5] | (2) CAR [−3, 5] | (3) CAR [0, 1] | (4) CAR [0, 1] | (5) CAR [−1, 1] | (6) CAR [−1, 1] | (7) CAR [−1, 3] | (8) CAR [−1, 3] | (9) CAR [−3, 10] | (10) CAR [−3, 10] |
|---|---|---|---|---|---|---|---|---|---|---|
| log(*NEWS PRESSURE UNCORRECTED*) | 7.883*** | 7.264* | 3.792* | 4.570 | 5.483** | 5.316* | 5.482* | 4.725 | 7.787** | 8.690$^{+}$ |
| | (2.016) | (3.206) | (1.625) | (2.692) | (1.779) | (2.558) | (2.238) | (2.978) | (2.694) | (4.247) |
| log(*EBIT*) | | −0.345 | | −0.212 | | −0.190 | | −0.148 | | 0.379 |
| | | (0.440) | | (0.206) | | (0.210) | | (0.319) | | (0.589) |
| log(*EMPLOYEES*) | | −0.227 | | 0.069 | | −0.156 | | −0.277 | | −0.501 |
| | | (0.331) | | (0.167) | | (0.178) | | (0.221) | | (0.556) |
| log(*CAPEX*) | | −0.616 | | 0.012 | | −0.132 | | −0.440 | | −0.740 |
| | | (0.503) | | (0.186) | | (0.240) | | (0.312) | | (0.591) |
| log(*OPEX*) | | 1.384 | | 0.034 | | 0.469$^{+}$ | | 0.939 | | 1.044 |
| | | (0.962) | | (0.303) | | (0.272) | | (0.605) | | (1.354) |
| *IT_INDUSTRY* | | 0.326 | | −0.946 | | −1.093 | | −0.460 | | 0.744 |
| | | (0.910) | | (0.664) | | (0.654) | | (0.717) | | (1.267) |
| Intercept ($\beta_0$) | −4.913*** | −9.222$^{+}$ | −2.234* | −1.254 | −3.394*** | −4.328** | −3.287** | −5.455 | −5.586** | −11.313 |
| | (0.891) | (4.807) | (0.891) | (1.560) | (0.802) | (1.504) | (0.931) | (3.369) | (1.709) | (7.335) |
| Time fixed effects | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Model | MM | MM | MM | MM | MM | MM | MM | MM | MM | MM |
| $N$ | 311 | 208 | 311 | 208 | 311 | 208 | 311 | 208 | 311 | 208 |
| $R^2$ | 0.045 | 0.081 | 0.060 | 0.122 | 0.070 | 0.146 | 0.050 | 0.090 | 0.025 | 0.061 |

*Notes.* OLS estimates; $N$ is in firms. MM, market model. Time fixed effects are dummies for year and day of the week. Standard errors are in parentheses.

$^{+}$, *, **, and *** indicate significance at the 10%, 5%, 1%, and 0.1% levels, respectively.

**Figure 6.** Predicted Cumulative Abnormal Returns (in Percent) and News Pressure




*Notes*. The figure displays the predicted CARs conditional on different levels of news pressure. Panels A and B show CARs for corrected news pressure; panels C and D, uncorrected news pressure. Whiskers give 95% confidence intervals.

the counterfactual returns. The FF3 model accounts for systematic divergence between two classes of stocks that tended to outperform markets—namely, stocks of firms with lower market capitalization (i.e., small caps) and stocks of firms with high book-to-market ratios (i.e., value stocks). The FFC4 model adjusts additionally for stock momentum based on continuing tendencies in the stock price. In Table 11, columns (1) and (2) summarize the CARs for the event window [−3, 5] in the FF3 and FFC4 models. Moreover, the results hold when extending the estimation window to 250 trading days, as is reported in column (3). The full results are reported in Online Appendix C, Section 2. Overall, we conclude that our results are consistent for these alternative models.

Second, we assess sensitivity toward subsamples and alternative operationalizations of news pressure as in Study 1. Continuing with Table 11, column (4) excludes firms in the IT industry, and column (5) excludes firms in the finance and insurance industry (*FIN_INDUSTRY*) to assess whether the results are driven by those industries that are arguably more data intensive. The results are consistent with the baseline. Column (6) shows that the results are qualitatively similar when excluding Friday announcements.

Third, identical to Study 1, we test different operationalizations of news pressure, including untransformed, character count, and word count (see Online Appendix C, Section 2). We also exclude extreme observations in

**Table 11.** Summary of Event Study Robustness Checks

| | CAR [−3, 5] | | | | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| | FF3 model | FFC4 model | Extended estimation window | Without IT | Without finance | Without Fridays |
| log(*NEWS PRESSURE*) | 6.376* | 6.865* | 7.126** | 7.952*** | 9.158** | 7.811** |
| | (2.364) | (2.770) | (2.132) | (2.030) | (2.510) | (2.119) |
| Intercept ($\beta_0$) | −3.991*** | −4.700*** | −4.585** | −5.307*** | −5.087*** | −4.255** |
| | (0.826) | (0.860) | (1.318) | (0.769) | (1.262) | (1.525) |
| Time fixed effects | Yes | Yes | Yes | Yes | Yes | Yes |
| Model | FF3 | FFC4 | MM | MM | MM | MM |
| *N* | 311 | 311 | 311 | 271 | 251 | 249 |

*Notes*. OLS estimates; *N* is in firms. Time fixed effects are dummies for year and day of the week. MM, market model; FF3, Fama–French three-factor model; FFC4, Fama–French–Carhart four-factor model. Standard errors are in parentheses.

*, **, and *** indicate significance at the 5%, 1%, and 0.1% levels, respectively.

CARs and extreme breach severity, and we control explicitly for the number of breached records. The results are consistent. We also include additional month fixed effects and find the results confirmed.[27]

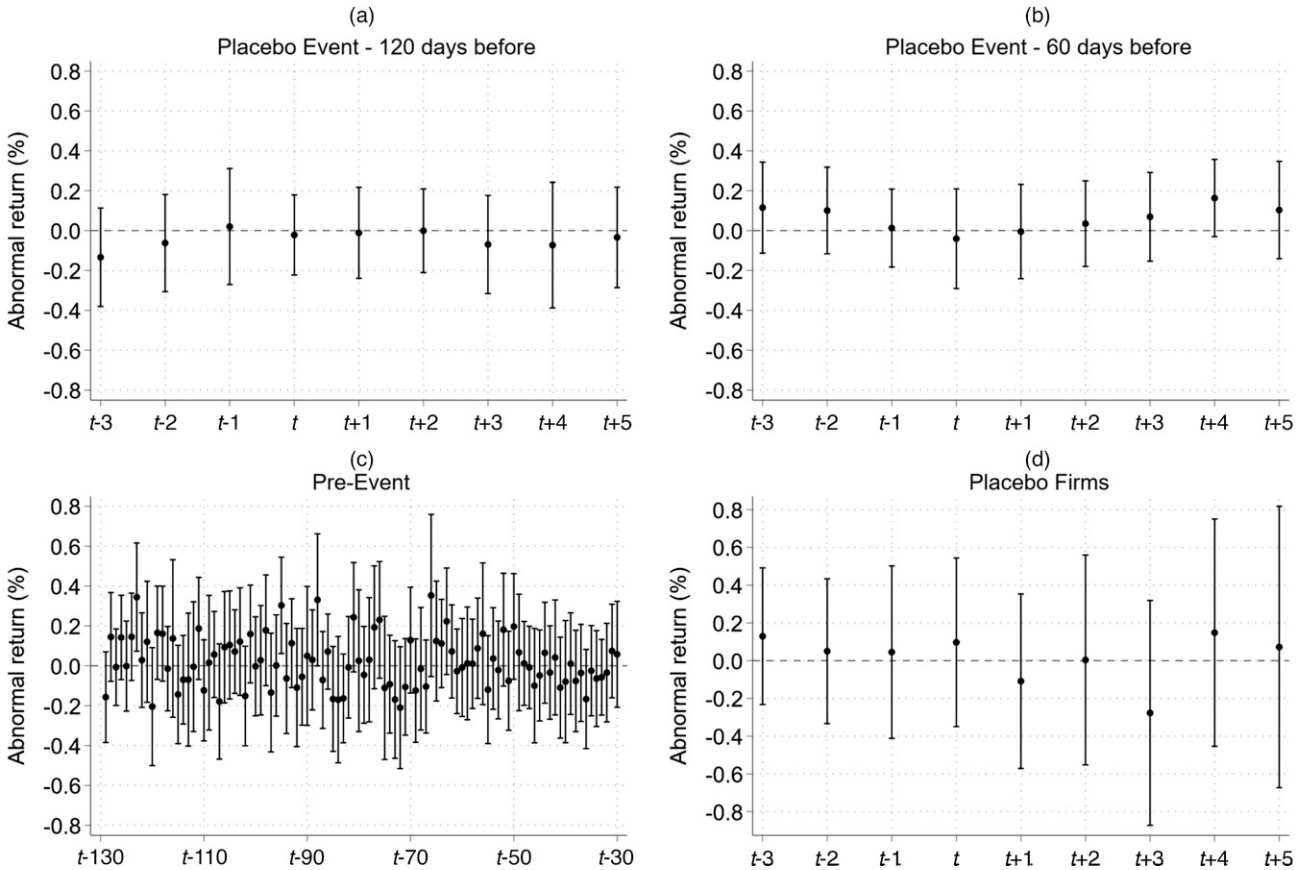
Fourth, Figure 7 summarizes a series of falsification tests (e.g., similar to Grewal et al. 2019). Concern may exist that the observed CARs are just due to random chance (e.g., because of confounding events, composition of the sample). In a first test, we estimate market reactions to placebo events in the pre-event period. We rely on the identical parameters but place the event window 120 days and 60 days before the actual data breach announcement in panels A and B, respectively. We adjust the estimation window accordingly. Panels A and B plot the ARs over the placebo event window, which are close to 0 and all insignificant. In a complementary test, we seek to assess pre-event abnormal returns. We therefore define the estimation period as event window. We move the estimation window accordingly. Panel C plots the abnormal returns over this artificial event window. Before the actual event window, the estimated ARs are close to 0. On only two days long before the event window, we observe significant ARs; however, they show the opposite direction (i.e., they are positive). Both observations lend confidence that our findings are specific to the event window around the data breach announcement and not an artifact of preexisting abnormal returns. In a final test, we leave the event dates unchanged but assign a random sample of not breach-announcing firms to these dates (e.g., Foerderer 2020). Panel D plots the resulting ARs. All of them are insignificant, as expected, and the confidence intervals are wide. This observation supports that our results are not an artifact of the event dates but specific to the firms that announce a data breach.

Fifth, we explore long-term abnormal returns using the buy-and-hold abnormal return (BHAR) method (Barber and Lyon 1997, Kothari and Warner 1997). BHAR reflects the abnormal returns realized by buying stocks of the event-affected firms at the beginning of the event window and selling them at the end of the event window, compared with returns that could have been realized from a benchmark group of stocks over the same period. Benchmark securities are chosen based on market capitalization deciles and book-to-market ratios. Table 12 reports the results for a window of one year. Column (1) reports the estimates from an equal-weighted approach with rebalancing. Column (2) reports the results from a value-weighted approach with rebalancing. Columns (3) and (4) proceed in an identical manner but report estimates from a model without rebalancing. Overall, the results are consistent with the short-term analysis.[28] Section 3 in Online Appendix C provides the full technical details, the parameters, and the estimated equation, and it plots the predicted BHARs for interpretation.

Sixth, concern may exist that the results are spurious as a result of potential simultaneity between news pressure and attention; in other words, on days with high news pressure, more "eyeballs" are paying close attention, which should render strategic timing ineffective. Although our event study suggests the opposite, we conduct an additional analysis to address this concern. In the spirit of Da et al. (2011), we estimate Google search volume for the ticker symbols of the breached firms (e.g., "AAPL" for Apple Computer and "MSFT" for Microsoft). The idea is to estimate an abnormal search volume index (ASVI), which is defined as the search interest in the ticker of firm $i$ around the event window compared with the normalized median search interest for that firm's ticker during the prior weeks.

Table 13 reports the results. Breach-announcing firms show significant abnormal search interest during the event window; higher levels of news pressure on the announcement day are associated with lower abnormal search interest. Thus, Google search attention

**Figure 7.** Pre-Event Heterogeneity and Placebo Event




*Notes.* The figure plots the abnormal returns together with 95% confidence intervals. Panel A displays the ARs for a placebo event 120 days before the actual data breach announcement. Panel B is identical but for a placebo event 60 days before the announcement. Panel C plots the ARs for the estimation window. Panel D reports the ARs over the actual event window but for randomly selected placebo firms.

toward breached firms is significantly lower when the level of news pressure on the announcement day is higher. Although this analysis entails its own caveats, it contradicts the assertion that on busy days, more "eyeballs" are watching. Section 4 in Online Appendix C reports the full details. Complementary to the analysis of abnormal search interest, we examine the sentiments of tweets related to a breached firm during the event window. The analysis is reported in Online Appendix C, Section 5. In line, we find that negative tweet

**Table 12.** Long-Term Buy-and-Hold-Abnormal-Returns (in Percent) (One Year)

| | BHAR | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| log(*NEWS PRESSURE*) | 7.793* | 7.619* | 7.918* | 7.922* |
| | (3.308) | (3.266) | (3.354) | (3.352) |
| Intercept ($\beta_0$) | −15.634* | −12.713* | −11.618$^+$ | −11.597$^+$ |
| | (6.559) | (6.125) | (6.618) | (6.604) |
| Time fixed effects | Yes | Yes | Yes | Yes |
| Weighting | Equally | Value | Equally | Value |
| Rebalanced | Yes | Yes | No | No |
| *N* | 309 | 309 | 309 | 309 |
| $R^2$ | 0.070 | 0.067 | 0.068 | 0.068 |

*Notes.* OLS estimates; *N* differs from the short-term event study because of the availability of stock returns and variables for the extended event window. Estimations are based on the firm-level data set. Standard errors are in parentheses.
$^+$ and * indicate significance at the 10% and 5%, levels, respectively.

**Table 13.** Abnormal Ticker Search Volume

| | (1) ASVI [−3, 5] | (2) ASVI [0] | (3) ASVI [0, 1] | (4) ASVI [−1, 1] | (5) ASVI [−2, 2] |
|---|---|---|---|---|---|
| log(*NEWS PRESSURE*) | −0.138* | −0.090* | −0.138** | −0.167** | −0.167** |
| | (0.056) | (0.040) | (0.045) | (0.051) | (0.051) |
| Intercept ($\beta_0$) | 0.458*** | 0.298*** | 0.363*** | 0.386*** | 0.414*** |
| | (0.027) | (0.023) | (0.023) | (0.025) | (0.025) |
| Time fixed effects | Yes | Yes | Yes | Yes | Yes |
| *N* | 473 | 473 | 473 | 473 | 473 |

*Notes*. OLS estimates. The dependent variable is the ASVI from Google Trends, with values in brackets denoting the number of days around the data breach announcement. Standard errors are in parentheses.

*, **, and *** indicate significance at the 5%, 1%, and 0.1% levels, respectively.

sentiment is less pronounced when the announcement is on days with high news pressure.

# 7. Discussion

## 7.1. Summary of Findings

Our paper reports the results from two complementary empirical analyses to assess strategic timing as one potential explanation for markets' unresponsiveness to data breach announcements. In Study 1, we find evidence that data breach announcements coincide with predictably busy days in the media, which we interpret as evidence of strategic timing. In Study 2, we observe negative market reactions to data breach announcements and that higher levels of news pressure attenuate negative market reactions. Taken together, our findings support strategic timing as partial explanation of attenuated stock market reactions to data breach announcements.

## 7.2. Theoretical Contributions and Implications

Our findings offer three key contributions. First, our findings shed light on the apparent unresponsiveness of markets to data breach announcements. Extant literature is ambiguous about the economic effects of data breach announcements, as several studies find conflicting evidence regarding the direction, persistence, and magnitude of stock market reactions (e.g., Kannan et al. 2007, Goel and Shawky 2009, Gatzlaff and McCullough 2010, Morse et al. 2011, and Martin et al. 2017). Whereas prior work had noticed the market's unresponsiveness (e.g., Kvochko and Pant 2015 and Kwon and Johnson 2015), our study provides evidence of a fresh explanation that links attenuated market reactions to firms' strategic behavior with regard to the announcement timing.

Second, our findings also contribute novel insights to the literature on firm behavior after data breaches (Goode et al. 2017, Gwebu et al. 2018, Bansal 2019). Extant work has focused predominantly on firms' recovery behaviors, such as public apologies and consumer compensation campaigns. Our study suggests a further breach-related behavior by firms, strategic timing, and that it can influence the outcomes of data breach announcements. We also present evidence that firms' incentives for this strategic behavior differs depending on breach severity and cause, firm type, and type of data leaked.

Third, our findings contribute to research on the effectiveness of data breach regulations (Laube and Böhme 2016, Park 2019, Zou et al. 2019). Our study suggests that firms may—through strategic timing—evade some of the economic consequences that are expected to ensue when they are forced to publicly disclose data breaches. Because existing legislation is rooted, at least partly, in the notion that poor security practices will be deterred by these economic consequences (Romanosky et al. 2011 and Bisogni 2016), this finding is relevant, as it suggests that firms might act strategically to prevent negative consequences. Our study adds to prevailing concerns that existing notification laws might be ineffective (Laube and Böhme 2016; Park 2019; Zou et al. 2018, 2019).

Finally, our study contributes to the discourse on whether firms time the announcements of bad news (e.g., Patell and Wolfson 1982, Damodaran 1989, Hong and Stein 1999, DellaVigna and Pollet 2009, and Doyle and Magilke 2009). Against the backdrop of recent evidence being mixed (Michaely et al. 2014, deHaan et al. 2015, Michaely and Rubin 2016), our study offers new empirical evidence that suggests that firms strategically time one type of negative news, data breach announcements, and by doing so can avoid negative market reactions. These findings are also adding to research on strategic timing in economics by helping us to understand the downstream market consequences of strategic timing by firms (Eisensee and Strömberg 2007, Garz and Sörensen 2017, Durante and Zhuravskaya 2018).

## 7.3. Practical Contributions

Our findings have practical implications for consumers, lawmakers, firms, and investors. From our analysis, there is evidence of firms strategically timing the announcement of data breaches to days of higher news pressure, which eventually can attenuate

negative market reactions. This is detrimental for consumers who rely on the media for timely information about a data breach and on market mechanisms that incentivize firms to protect consumer data. Perhaps most important, strategic timing is most prevalent for data breaches that are of greatest interest to consumers—namely, those that are more severe and involve healthcare and financial data as well as credentials. For lawmakers and regulators, our findings have implications for the design of data breach notification laws. Arguably, longer notification deadlines provide firms with leeway for hiding the data breach announcements in the press. Shorter deadlines—such as the 72-hour disclosure deadline mandated in the GDPR—could potentially restrict strategic timing. Nevertheless, potentially negative side effects need to be considered. For example, if a firm reports a breach before successful remediation, it risks allowing copycat hackers to exploit its existing vulnerabilities and impose additional harm (Mitra and Ransbotham 2015). Alternative approaches to avoid strategic timing and the negative side effects of shorter windows could require the announcement on specific dates. For firms, strategic timing represents a strategy that can reduce the negative market outfall of data breaches. For investors, our study suggests that assessing the market impact of data breach announcements requires an evaluation of the news pressure on the announcement day.

### 7.4. Interpretation, Limitations, and Future Research

It is important to consider our findings in light of the studies' boundary conditions and limitations. First, the observed results should be interpreted as evidence that data breach announcements coincide with busy days in the trade press and that this can attenuate investors' negative reactions—but not as definitive evidence of intentional behavior by firms. Still, the correlation between data breach announcements and busy days in the media is problematic because the result remains the same: attention is reduced. Second, the obtained estimates should be interpreted in light of the use of the *WSJ* as an indicator of news pressure. Our estimates should be understood to capture news pressure in the trade press but not necessarily in the wider general press, and if the goal is to infer general news pressure, one may anticipate measurement error. Third, our analysis of stock market reactions is limited by the fact that unobservable factors influenced the results. Although our regressions have adjusted for time fixed effects as well as various firm-specific controls, they are limited in capturing heterogeneity. Fourth, our findings do not permit conclusions about the time window within which firms can adjust their announcement because this would require an additional study in the spirit of a policy-change analysis. For example, one could evaluate strategic timing before and after the imposture of stricter notification requirements in a state. This is, however, beyond what can be tested with our data. Finally, one could argue that firms with data breach insurance need not engage in strategic timing, whereas firms with media consultants exhibit a stronger inclination toward strategic timing. However, our data cannot assess these questions.[29]

## 8. Conclusion

Motivated by markets' apparent unresponsiveness to data breach announcements, we investigated whether firms' strategic behavior in the timing of breach announcements could explain attenuated stock market reactions. Through two studies, we find evidence of strategic timing. We observe that data breach announcements coincide with predictably busy days in the media. Strategic timing is more prevalent for severe than for nonsevere data breaches, for breaches of public rather than for private firms, for breaches stemming from firm-internal causes as opposed to firm-external causes, and for breaches involving healthcare and financial data as well as credentials. Leveraging an event study analysis, we observe that higher levels of news pressure on the announcement day attenuate negative market reactions to the data breach announcement. Taken together, our findings indicate that the markets' unresponsiveness can, at least partially, be explained by firms' strategic behavior regarding announcement timing.

### Acknowledgments

The authors are extremely grateful for the data provided by the Identity Theft Resource Center. The authors also express special thanks to department editor Chris Forman, the associate editor, and the anonymous reviewers, as well as participants of the 30th Workshop on Information Systems and Economics (WISE 2019). The authors also extend their gratitude to Alessandro Acquisti, Bernd Skiera, and Jason Thatcher for their helpful comments.

### Endnotes

[1] For example, in September 2018, Facebook reported a breach of approximately 50 million user records. The breach was reported in *Wired* and *TechCrunch,* yet it did not receive substantial public attention. The announcement was released late on a Friday when the media were also occupied by the Brett Kavanaugh hearings and the Securities and Exchange Commission controversy over Elon Musk (Ingram 2018).

[2] Researchers have been interested in understanding the consequences of data breach regulations on various consumer outcomes, including identity theft (e.g., Romanosky et al. 2011 and Laube and Böhme 2016). The findings are mixed. On the one hand, Romanosky et al. (2011) demonstrate that state-level data breach notification laws reduce identity theft. On the other hand, the model of Laube and Böhme (2016) implies that the enforcement of notification laws

is impossible because auditors cannot distinguish between concealment and nescience of firms.

[3] The distribution shows a bunching around the 30-day mark. Given that several notification laws impose a 30-day deadline, one may interpret this as a sign of data accuracy. We thank an anonymous reviewer for this interpretation.

[4] Section 4 in Online Appendix A evaluates alternative approaches for constructing a news pressure measure from *WSJ* data that we discarded.

[5] We thank an anonymous reviewer for this suggestion.

[6] Online Appendix Figure A1 further shows that there is substantial day-to-day variation in the variable.

[7] The weekend edition presents advertisements and a differently sized logo at the top of the front page. It also includes reviews, design, travel, and fashion articles. It is about 50 pages long, as opposed to the approximately 30 pages of the weekday issue. The order of news items within the What's News column is flipped; World News comes first, and Business & Finance follows. See Online Appendix A, Section 5.

[8] The accuracy rates are available in deHaan et al. (2015) until 2011.

[9] Specifically, deHaan et al. (2015) use four sources and keep earnings in the sample if they can be validated by at least two sources. Our data access is restricted to two widely used databases, Compustat and I/B/E/S.

[10] We thank an anonymous reviewer for this suggestion.

[11] On approximately 20.31% (79.25%) of the days in the full (public firms) sample, no data breach was announced, leaving variation for days without announcements.

[12] We use the ivreg2 package for the weak instrument tests (Baum et al. 2007). To interpret the Kleibergen–Paap rk Wald $F$-statistic, we follow Baum et al. (2007) and apply the critical values computed for the Cragg–Donald statistic.

[13] The negative coefficient on $\log(EARN\ ANNCMNTS)_{t\text{-}1}$ in column (2) of Table 3 shows that the more earnings announcements are released in $t-1$, the lower the news pressure in $t$. One may interpret this as refuting the concern that, on busy days, the *WSJ* postpones the coverage of data breach announcements to less busy days. If the *WSJ* would postpone, we would expect that the more earnings announcements are released in $t-1$, the higher the news pressure in $t$—and thus a positive coefficient. We thank an anonymous reviewer for this interpretation.

[14] We use the stdbeta package for obtaining this interpretation.

[15] For corrected news pressure, none of the lags or leads is significant. For uncorrected news pressure, the third lag is significant ($p < 0.05$). Section 9 in Online Appendix B reports the results for an extended window, which are consistent.

[16] This result suggests that the simple OLS estimate from column (3) underestimates the effect, consistent with the idea of the research design, which holds that news pressure consists of predictable and unpredictable components. The unpredictable component likely reduces the association between news pressure and data breach announcements.

[17] We use the stdbeta package for obtaining the estimate. Moreover, to allow for this interpretation, we assume that the value 0 in the variable means that zero records were announced to be breached. We impose this assumption for simplicity and caution against an overinterpretation.

[18] Multinomial logistic regression models extend logistic regressions to response variables that have more than two categories. Each category of the dependent variable is then estimated in an individual equation, and a test between the coefficients across the equations (i.e., a Wald test) is used to test for differences in the models (Greene 2012). This approach is conceptually equivalent to fitting a series of logistic regression models. The models include a simpler set of time fixed effects (i.e., day of the week and year) to avoid convergence issues.

[19] We also test how news pressure on preceding days influences the announcement of severe data breaches (see Online Appendix B, Section 2). We find evidence that news pressure on $t-1$ decreases the likelihood of a severe data breach being announced, but we find no influence for days $t-2$ and $t-3$. We thank an anonymous reviewer for this idea.

[20] This observation also contradicts the rival explanation that the *WSJ* may deliberately delay reporting severe breaches. We thank an anonymous reviewer for suggesting this interpretation.

[21] We thank an anonymous reviewer for proposing this test.

[22] We thank the review team for suggesting to explore state-level heterogeneity. This analysis should be interpreted in light of two challenges: (1) it faces the challenge of capturing unrelated heterogeneity between states, and (2) it relies on a simplified classification of notification deadlines.

[23] Section 8 in Online Appendix B reports an additional robustness check in which we drop data breach announcements that were announced immediately after their discovery.

[24] We rely on a cross-sectional event study for three reasons. First, we chose an event study approach over other research designs to align with extant research that has investigated the market reactions to data breach announcements (see Section 2 in Online Appendix A, which lists the existing studies that used this method). Second, although panel data could theoretically provide greater information, their use is challenged by data availability. Although the stock market returns are available on a daily base, most firm-level controls are available only annually, constraining the additional information that could at all enter the regression. Finally, one of the key advantages of panel data—firm-level fixed-effects—would not be realizable in our case: including firm fixed effects would discard all firms that experience only one data breach (i.e., most of the sample). Even if a large-enough sample of firms were available, it would be difficult to interpret the resulting estimates, as firms that experienced multiple data breaches could represent a biased sample as a result of not taking appropriate countermeasures after the first data breach.

[25] An alternative approach would be to exploit that some firms announce several breaches over time, which would permit the inclusion of firm fixed effects. We discarded this approach for three reasons. First, it is only a handful of firms that experience multiple breaches. Second, even if feasible, the resulting estimates could be biased by serial correlation. Last, such an approach would also be challenged by overlaps in the estimation and event windows. We thank an anonymous reviewer for this suggestion.

[26] The plots show CARs winsorized at the 5th and 95th percentiles of their empirical distribution to ease visual interpretation.

[27] The estimates are consistent in terms of their direction. For the model including day-of-week, week, month, and year fixed effects, several coefficients drop below the significance level. However, note that in this model the degrees of freedom are small.

[28] The coefficient on (standardized) *NEWS PRESSURE* is consistently positive. The intercept is consistently negative. The effects of news pressure on the announcement day have persistent market reactions, and the persistent effects are more marked than the short-term results.

[29] We thank an anonymous reviewer for this suggestion.

## References

Acquisti A, Brandimarte L, Loewenstein G (2015) Privacy and human behavior in the age of information. *Sci.* 347(6221):509–514.

Acquisti A, Friedman A, Telang R (2006) Is there a cost to privacy breaches? An event study. *Proc. 27th Internat. Conf. Inform. Systems* (Association for Information Systems, Atlanta).

Agility PR Solutions (2020) Top 10 U.S. newspapers by circulation. Accessed September 10, 2020, https://www.agilitypr.com/resources/top-media-outlets/top-10-daily-american-newspapers/.

Angrist JD, Imbens GW, Rubin DB (1996) Identification of causal effects using instrumental variables. *J. Amer. Statist. Assoc.* 91(434): 444–455.

Bansal G (2019) Restoring trust after an insider breach: Both the genders matter—CEOs and users. *J. Comput. Inform. Systems* 61(1): 11–29.

Barber BM, Lyon JD (1997) Detecting long-run abnormal stock returns: The empirical power and specification of test statistics. *J. Financial Econom.* 43(3):341–372.

Baum CF, Schaer ME, Stillman S (2007) Enhanced routines for instrumental variables/generalized method of moments estimation and testing. *Stata J.* 7(4):465–506.

Beaver WH (1968) The information content of annual earnings announcements. *J. Accounting Res.* 6(1):67–92.

Bisogni F (2016) Proving limits of state data breach notification laws: Is a federal law the most adequate solution? *J. Inform. Policy* 6:154–205.

Borah A, Tellis GJ (2014) Make, buy, or ally? Choice of and payoff from announcements of alternate strategies for innovations. *Marketing Sci.* 33(1):114–133.

Bose I, Leung ACM (2019) Adoption of identity theft countermeasures and its short-and long-term impact on firm value. *MIS Quart.* 43(1):313–327.

Brown SJ, Warner JB (1985) Using daily stock returns: The case of event studies. *J. Financial Econom.* 14(1):3–31.

Campbell K, Gordon LA, Loeb MP, Zhou L (2003) The economic cost of publicly announced information security breaches: Empirical evidence from the stock market. *J. Comput. Security* 11(3): 431–448.

Caroll CE, McCombs ME (2003) Agenda setting effects of business news on the public's images and opinions about major corporations. *Corporate Reputation Rev.* 6(1):36–46.

Cavusoglu H, Mishra B, Raghunathan S (2004) The effect of internet security breach announcements on market value: Capital market reactions for breached firms and internet security developers. *Internat. J. Electronic Commerce* 9(1):69–104.

Chambers AE, Penman SH (1984) Timeliness of reporting and the stock price reaction to earnings announcements. *J. Accounting Res.* 22(1):21–47.

Da Z, Engelberg J, Gao P (2011) In search of attention. *J. Finance* 66(5):1461–1499.

Damodaran A (1989) The weekend effect in information releases: A study of earnings and dividend announcements. *Rev. Financial Stud.* 2(4):607–623.

deHaan E, Shevlin T, Thornock J (2015) Market (in)attention and the strategic scheduling and timing of earnings announcements. *J. Accounting Econom.* 60(1):36–55.

DellaVigna S, Pollet JM (2009) Investor inattention and Friday earnings announcements. *J. Finance* 64(2):709–749.

Doyle JT, Magilke MJ (2009) The timing of earnings announcements: An examination of the strategic disclosure hypothesis. *Accounting Rev.* 84(1):157–182.

Durante R, Zhuravskaya E (2018) Attack when the world is not watching? US news and the Israeli-Palestinian conflict. *J. Political Econom.* 126(3):1085–1133.

Eisensee T, Strömberg D (2007) News droughts, news floods, and US disaster relief. *Quart. J. Econom.* 122(2):693–728.

Foerderer J (2020) Interfirm exchange and innovation in platform ecosystems: Evidence from Apple's Worldwide Developers Conference. *Management Sci.* 66(10):4772–4787.

Foerderer J, Kude T, Mithas S, Heinzl A (2018) Does platform owner's entry crowd out innovation? Evidence from Google Photos. *Inform. Systems Res.* 29(2):444–460.

Garz M, Sörensen J (2017) Politicians under investigation: The news media's effect on the likelihood of resignation. *J. Public Econom.* 153(September):82–91.

Gatzlaff KM, McCullough KA (2010) The effect of data breaches on shareholder wealth. *Risk Management Insurance Rev.* 13(1): 61–83.

Goel S, Shawky HA (2009) Estimating the market impact of security breach announcements on firm values. *Inform. Management* 46(7):404–410.

Goldstein J, Chernobai A, Benaroch M (2011) An event study analysis of the economic impact of IT operational risk and its subcategories. *J. Assoc. Inform. Systems* 12(9):606–631.

Goode S, Hoehle H, Venkatesh V, Brown SA (2017) User compensation as data breach recovery action: An investigation of the Sony PlayStation network breach. *MIS Quart.* 41(3):703–727.

Graham JR, Harvey CR, Rajgopal S (2005) The economic implications of corporate finance reporting. *J. Accounting Econom.* 40(1–3):3–73.

Greene WH (2012) *Econometric Analysis*, 7th ed. (Pearson Education, Upper Saddle River, NJ).

Grewal J, Riedl EJ, Serafeim G (2019) Market reaction to mandatory nonfinancial disclosure. *Management Sci.* 65(7): 3061–3084.

Gwebu KL, Wang J, Wang L (2018) The role of corporate reputation and crisis response strategies in data breach management. *J. Management Inform. Systems* 35(2):683–714.

Han K, Mithas S (2013) Information technology outsourcing and non-IT operating costs: An empirical investigation. *MIS Quart.* 37(1):315–331.

Hirshleifer D, Lim SS, Theo SH (2009) Driven to distraction: Extraneous events and underreaction to earning news. *J. Finance* 64(5):2289–2325.

Hong H, Stein JC (1999) A unified theory of underreaction, momentum trading, and overreaction in asset markets. *J. Finance* 54(6): 2143–2184.

Identity Theft Resource Center (ITRC) (2019) 2019 end-of-year data breach report. Accessed September 10, 2020, https://www.idtheftcenter.org/wp-content/uploads/2020/01/01.28.2020_ITRC_2019-End-of-Year-Data-Breach-Report_FINAL_Highres-Appendix.pdf.

Identity Theft Resource Center (ITRC) (2020) 2020 year in review data breach report. Accessed February 8, 2021, https://notified.idtheftcenter.org.

Ingram M (2018) Are we all suffering from data breach fatigue? *Columbia Journalism Rev.* (October 3), https://www.cjr.org/the_media_today/facebook-data-breach.php.

IT Governance (2019) Data breach notification laws by state: Compliance. Accessed August 30, 2019, https://www.itgovernanceusa.com/data-breach-notification-laws#C1.

Kaniel R, Parham R (2017) WSJ Category Kings—The impact of media attention on consumer and mutual fund investment decisions. *J. Financial Econom.* 123(2):337–356.

Kannan K, Rees J, Sridhar S (2007) Market reactions to information security breach announcements: An empirical analysis. *Internat. J. Electronic Commerce* 12(1):69–91.

Ko M, Dorantes C (2006) The impact of information security breaches on financial performance of the breached firms: An empirical investigation. *J. Inform. Tech. Management* 17(2):13–22.

Kohli R, Devaraj S, Ow TT (2012) Does information technology investment influence a firm's market value? A case of

non-publicly traded healthcare firms. *MIS Quart.* 36(4): 1145–1163.

Kothari S, Warner JB (1997) Measuring long-horizon security price performance. *J. Financial Econom.* 43(3):301–339.

Kvochko E, Pant R (2015) Why data breaches don't hurt stock prices. *Harvard Bus. Rev.* (March 31), https://hbr.org/2015/03/why-data-breaches-dont-hurt-stock-prices.

Kwon J, Johnson ME (2015) The market effect of healthcare security: Do patients care about data breaches? Workshop on the Economics of Information Security (WEIS), www.econinfosec.org/archive/weis2015/papers/WEIS_2015_kwon.pdf.

Laube S, Böhme R (2016) The economics of mandatory security breach reporting to authorities. *J. Cybersecurity* 2(1):29–41.

Lin M, Lucas HC Jr, Shmueli G (2013) Too big to fail: Large samples and the *p*-value problem. *Inform. Systems Res.* 24(4):906–917.

Malhotra A, Malhotra CK (2011) Evaluating customer information breaches as service failures: An event study approach. *J. Service Res.* 14(14):44–59.

Mani D, Barua A, Whinston AB (2013) Outsourcing contracts and equity prices. *Inform. Systems Res.* 24(4):1028–1049.

Martin KD, Borah A, Palmatier RW (2017) Data privacy: Effects on customer and firm performance. *J. Marketing* 81(1):36–58.

McCombs ME, Shaw DL (1972) The agenda-setting function of mass media. *Public Opinion Quart.* 36(2):176–187.

McCombs ME, Shaw DL (1993) The evolution of agenda-setting research: Twenty-five years in the marketplace of ideas. *J. Comm.* 43(2):58–67.

Michaely R, Rubin A (2016) Are Friday announcements special? Overcoming selection bias. *J. Financial Econom.* 122(1):65–85.

Michaely R, Rubin A, Vedrashko A (2014) Corporate governance and the timing of earning announcements. *Rev. Finance* 18(6): 2003–2044.

Miller AR, Tucker C (2009) Privacy protection and technology diffusion: The case of electronic medical records. *Management Sci.* 55(7):1077–1093.

Mitra S, Ransbotham S (2015) Information disclosure and the diffusion of information security attacks. *Inform. Systems Res.* 26(3): 565–584.

Morse EA, Raval V, Wingender JRJ (2011) Market price effects of data security breaches. *Inform. Security J.* 20(6):263–273.

Park S (2019) Why information security law has been ineffective in addressing security vulnerabilities: Evidence from California data breach notifications and relevant court and government records. *Internat. Rev. Law Econom.* 58(June):132–145.

Patell JM, Wolfson MA (1982) Good news, bad news, and the intraday timing of corporate disclosures. *Accounting Rev.* 57(3):509–527.

Romanosky S, Hoffman D, Acquisti A (2014) Empirical analysis of data breach litigation. *J. Empirical Legal Stud.* 11(1):74–104.

Romanosky S, Telang R, Acquisti A (2011) Do data breach disclosure laws reduce identity theft? *J. Policy Anal. Management* 30(2):256–286.

Schwartz PM, Janger EJ (2007) Notification of data security breaches. *Michigan Law Rev.* 105(58):913–985.

Sorescu A, Warren N, Ertekin L (2017) Event study methodology in the marketing literature: An overview. *J. Acad. Marketing Sci.* 45(2):186–207.

Spanos G, Angelis L (2016) The impact of information security events to the stock market: A systematic literature review. *Comput. Security* 58(May):216–229.

Stock J, Yogo M (2005) Asymptotic distributions of instrumental variables statistics with many instruments. Andrews DWK, Stock JH, eds. *Identification and Inference for Econometric Models: Essays in Honor of Thomas Rothenberg* (Cambridge University Press, New York), 109–120.

Telang R, Wattal S (2007) An empirical analysis of the impact of software vulnerability announcements on firm stock price. *IEEE Trans. Software Engrg.* 33(8):544–557.

Weaver DH (2007) Thoughts on agenda setting, framing, and priming. *J. Comm.* 57(1):142–147.

Zou Y, Mhaidli AH, McCall A, Schaub F (2018) "I've got nothing to lose": Consumers' risk perceptions and protective actions after the Equifax data breach. *USENIX Symposium on Usable Privacy and Security (SOUPS)*, Baltimore, MD, 197–216.

Zou Y, Danino S, Sun K, Schaub F (2019) You "might" be affected: An empirical analysis of readability and usability issues in data breach notifications. *CHI 2019*, Glasgow, Scotland, UK, 1–9.

Copyright 2022, by INFORMS, all rights reserved. Copyright of Management Science is the property of INFORMS: Institute for Operations Research and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.